

**Design of a low-power Fractional-N Phase Locked Loop using a
Differential Ring-VCO in 45-nm CMOS**

A PROJECT REPORT

Submitted By

BL.EN.U4ECE22218

Hanu Sharma

BL.EN.U4ECE22219

J Sai Kesav

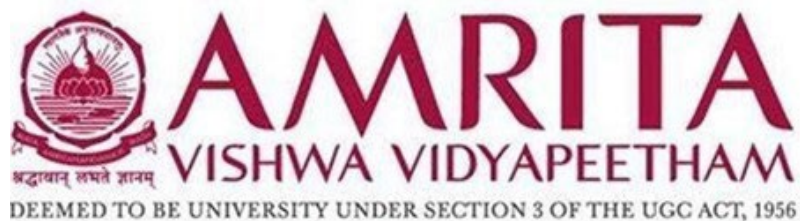
In partial fulfilment for the award of the degree

Of

BACHELOR OF TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING



AMRITA SCHOOL OF ENGINEERING, BENGALURU

AMRITA VISHWA VIDYAPEETHAM

BENGALURU, 560 035

May 2026



BONAFIDE CERTIFICATE

This is to certify that the project report entitled “**Design of a low-power Fractional-N Phase Locked Loop using a Differential Ring-VCO in 45-nm CMOS**” submitted by

BL.EN.U4ECE22218

Hanu Sharma

BL.EN.U4ECE22219

J Sai Kesav

in partial fulfilment of the requirements for the award of the Degree Bachelor of Technology in “Electronics and Communication Engineering” is a Bonafide record of the work carried out under my(our) guidance and supervision at Amrita School of Engineering, Bengaluru.

Mr. Vignesh V

Assistant Professor (Sr. Gr.)
Department of ECE
Amrita School of Engineering
Bengaluru

Dr. T. K. Ramesh

Chairperson
Department of ECE
Amrita School of Engineering
Bengaluru

This project report was evaluated by us on _____ .

Examiner 1
(Ms. Sonali Agrawal)

Examiner 2
(Mr. Swaminadhan R.)

ACKNOWLEDGEMENT

We extend our deepest gratitude to the divine forces that govern all endeavours for their blessings that have illuminated our path and guided us through the successful completion of this project. Our sincere pranams are offered at the lotus feet of the Universal Guru, **Mata Amritanandamayi Devi**, Chancellor of Amrita Vishwa Vidyapeetham, whose boundless inspiration and unwavering support have been pivotal in shaping our educational pursuits, including this project.

We express our sincere gratitude to **Dr. Manoj P.**, the esteemed director of Amrita University, Bengaluru. Dr. Manoj P's visionary leadership and continuous encouragement have played a pivotal role in fostering an environment of academic excellence and innovation at our institution. His unwavering commitment to the pursuit of knowledge and technological advancements has been a guiding force for us throughout this project.

A special acknowledgement is reserved for **Dr. Sriram Devanathan**, Principal, Amrita School of Engineering, Bengaluru, whose unwavering support and motivational guidance have been instrumental in propelling us towards excellence in our project. His leadership has been a beacon, inspiring us to reach new heights.

Our gratitude extends to **Dr. T. K. Ramesh**, the respected Chairman of the Department of ECE, for his expertise in the field, which has played a crucial role in steering us towards the successful completion of this project.

Our gratitude extends to our project advisor, **Mr. Vignesh V.**, Assistant Professor S. G., for his singular guidance and expertise in VLSI. His patient guidance, unwavering support, and valuable suggestions have been pivotal in shaping the trajectory of our project.

Special thanks are also extended to the project evaluators, **Ms. Sonali Agrawal** and **Mr. Swaminadhan R**, Assistant Professors of the Department of Electronics and Communication Engineering, for their valuable insights and expertise in the field of VLSI.

Our appreciation also extends to the dedicated staff members of the Department of Electronics and Communication Engineering. Their constant support, encouragement, and guidance have fostered a nurturing and intellectually stimulating environment.

ABSTRACT

This project presents the design and analysis of a CMOS Fractional-N Phase-Locked Loop (PLL) designed for wideband wireless applications. Implemented in a 45nm technology using Cadence Virtuoso, the design focuses on core sub-blocks including a zero dead-zone Phase Frequency Detector (PFD), a wide-swing cascode Charge Pump (CP), a ring-based Voltage-Controlled Oscillator (VCO), a Multi-Modulus Frequency Divider (MMFD), and a third-order MASH 1-1-1 Sigma-Delta Modulator. The architecture aims to achieve fine frequency resolution, low phase noise, and a wide tuning range to meet the demands of modern communication standards like 5G and Wi-Fi. Individual block functionality was successfully verified through simulation, establishing a foundation for a fully integrated, high-performance frequency synthesizer capable of supporting advanced wireless systems.

TABLE OF CONTENTS

Acknowledgement	iii
Abstract	iv
CHAPTER 1: INTRODUCTION	1
1.1 Introduction	1
1.2 Objectives	4
1.3 Methodology	5
1.4 Outcome(s)	6
1.5 Time Plan	7
1.6 Contents of The Report	7
CHAPTER 2: STATE OF ART	8
2.1 Introduction	8
2.2 Related Literature	8
2.3 Summary of Literature Survey	15
CHAPTER 3: DESIGN AND ANALYSIS	27
3.1 Introduction	27
3.2 Phase Frequency Detector and Charge Pump	27
3.3 Voltage Controlled Oscillator (VCO)	30
3.4 MMFD	34
3.5 MASH 1-1-1	38
3.6 Fractional N-PLL	44
CHAPTER 4: RESULT AND DISCUSSIONS	45
4.1 Introduction	45
4.2 Phase Frequency Detector and Charge Pump Results	45
4.3 Voltage Controlled Oscillator Results	46
4.4 Multi-Modulus Frequency Divider Results	47
4.5 Sigma Delta Modulator Results	49

4.6 PLL Results	51
4.7 Comparison table	54
CHAPTER 5: CONCLUSION AND FUTURE WORK	55
5.1 Conclusion	55
5.2 Future Work	55
CHAPTER 6: REFERENCES	56

CHAPTER 1: INTRODUCTION

1.1 Introduction

Modern communication systems, high-performance digital processors, and mixed-signal integrated circuits rely heavily on accurate and flexible frequency synthesis to generate the stable clock and carrier signals necessary for their operation. As wireless standards evolve, ranging from LTE and 5G transceivers to low-power IoT radios, there has been a rapid increase in the demand for frequency synthesizers that offer low phase noise, fine frequency resolution, wide tuning ranges, and fast settling times. In these systems, any instability or noise in the frequency source directly affects data throughput, spectral efficiency, and system reliability. A phase-locked loop (PLL) is a fundamental electronic control system used in tuning range of frequency. It is widely used in communication, clock generation, frequency synthesis and so on. PLL operates on the principle of feedback control to ensure that the phase and frequency of the output signal closely match those of a stable reference input signal [1]. The PLL has therefore become a fundamental building block in RF front ends, clock generators, microprocessors, and high-speed serial interfaces. A PLL achieves frequency synthesis by locking the phase of a controllable oscillator to a stable reference frequency. This closed-loop configuration ensures that the output frequency maintains long-term accuracy while preserving short-term stability.

1.1.1 Introduction to typical architectures

Traditional Integer-N PLLs restrict the feedback division ratio to whole numbers, meaning the output frequency is always an integer multiple of the reference. While this architecture is simple and widely used, it suffers from limited frequency resolution. To achieve fine steps, designers are forced to increase the reference frequency, but this leads to higher power consumption, stricter requirements on reference-clock generation, and difficulty meeting phase-noise specifications when using compact on-chip oscillators. As wireless standards demand channel spacing in the order of a few kilohertz or

less, Integer-N PLLs become impractical due to their inflexible architecture. The limited resolution also restricts synthesizer agility, making fast switching between adjacent channels extremely challenging.

Fractional-N PLLs address these limitations by allowing fractional division ratios in the feedback path. This is achieved by dynamically switching between multiple integer division values in such a way that the average division ratio generates the desired fraction for division. With the introduction of multi-modulus dividers and sigma-delta modulators, Fractional-N architectures achieve extremely fine frequency resolution without requiring high reference frequencies.

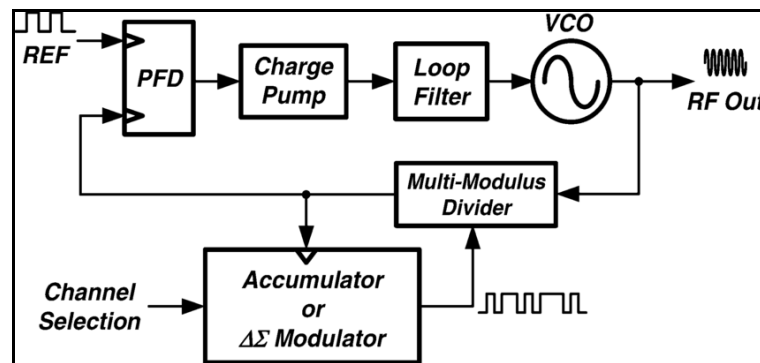


Fig 1.1 Fractional-N Phase Locked Loop

This is particularly useful in present day wireless standards where we require precise fractional frequencies for operation. This not only improves spectral efficiency but also reduces component stress, power consumption, and design complexity. However, traditional Fractional-N PLLs face their own challenges, such as quantization noise and fractional spurs. Modern designs employ higher-order sigma-delta modulators to shape this noise out of the loop bandwidth, significantly improving phase-noise performance. With the continuous development of integrated circuits, the size of the CMOS manufacturing process is decreasing and the supply voltage of circuit design is lowering. At the same time, the portability and small size of wireless communication equipment make its battery capacity decrease continuously [2]. Based on the above development trends, as the core module of wireless communication system, the PLL needs to achieve lower voltage, lower power consumption, low phase noise.

1.1.2 Present trends and applications

In the recent years, the rapid growth demand of applications in wired and wireless communications systems has led to increasing demand of the fully integrated, low-cost, low-power, and high-performance PLLs to provide accurate carrier signals that is phase aligned with the incoming high-precision reference clock signal [3]. In wideband wireless communication systems such as LTE-Advanced, 5G NR, Wi-Fi 6/7, and emerging ultra-wideband (UWB) technologies, the frequency synthesizer must support rapid hopping across broad spectrum allocations while maintaining extremely fine channel spacing and low phase noise. A Fractional-N PLL is particularly well-suited for these requirements because it offers high-resolution frequency tuning without demanding an excessively high reference frequency, enabling flexible generation of numerous carrier frequencies across wide bandwidths. This fine frequency granularity allows seamless support for variable channel bandwidths, modulation schemes, and carrier aggregation features that characterize modern wideband standards. Moreover, the use of sigma-delta modulation within the Fractional-N architecture helps suppress quantization noise and minimize fractional spurs, improving spectral purity, which is an essential requirement for minimizing adjacent-channel interference in crowded RF environments. As a result, Fractional-N PLLs have become the preferred synthesizer architecture in wideband RF transceivers, delivering the agility, noise performance, and frequency resolution necessary for next-generation wireless communication systems.

1.1.3 Design to overcome challenges

Despite their widespread use, traditional PLL architectures face several critical limitations when applied to modern wideband, multi-standard communication systems. Integer-N PLLs, with their fixed integer divider ratios, offer only coarse frequency resolution, forcing designers to increase the reference frequency to achieve smaller tuning steps, an approach that leads to higher power consumption, tougher phase-noise requirements, and difficulties in

integrating high-quality reference oscillators on chip. Even early Fractional-N PLLs, though offering finer resolution, struggled with problems such as fractional spurs, periodic jitter, and sensitivity to charge-pump mismatch. These spurs degrade spectral purity and can violate adjacent-channel power constraints in wireless systems. Additionally, limited noise-shaping capability in simpler modulators often resulted in quantization noise falling inside the loop bandwidth, directly impacting phase noise at the synthesizer output. As communication standards demand wider tuning ranges, lower noise, faster hopping, and tighter spectral masks, these shortcomings make traditional architectures increasingly inadequate, motivating the need for more advanced Fractional-N solutions with improved divider structures and sigma-delta modulation techniques.

Despite these advances, integrating a low-noise VCO, a linear charge pump, and a stable loop filter while ensuring minimal spur generation remains a major design challenge. The objective of this project is to address these challenges by implementing a robust Fractional-N PLL architecture consisting of a zero dead zone Phase Frequency Detector, a wide swing cascode Charge Pump, Loop Filter, low noise Voltage-Controlled Oscillator, Multi-Modulus Divider, and a third order Sigma-Delta Modulator. This design aims to achieve fine frequency resolution, fast settling, and improved phase-noise performance over traditional architectures, targeted at high frequency applications in wireless wideband systems.

1.2 Objectives

The primary objective of this project is to design and analyse a CMOS Fractional-N Phase-Locked Loop capable of generating high-resolution output frequencies suitable for modern high frequency wireless wideband systems. The tool used to implement this is Cadence Virtuoso and the technology being used is 45nm technology. The project aims to implement all core building blocks including the PFD, charge pump, loop filter, VCO, multi-modulus divider, and sigma-delta modulator and integrate them into a complete closed-loop architecture. Specific objectives include:

- Designing a VCO architecture for high frequency application that gives wide tuning range.
- Develop a charge pump and loop filter configuration for with zero dead zone and high output impedance respectively.
- Implement a high-resolution fractional frequency divider and sigma-delta modulator for fine frequency synthesis.
- To perform system-level simulations and transistor-level design verification to evaluate phase noise and frequency.

1.3 Methodology

The project follows a structured methodology beginning with a comprehensive literature review on PLL architectures and noise-shaping techniques. Following this, each subsystem is modelled and simulated individually to verify its standalone functionality. The Phase Frequency Detector and Charge Pump are first designed to provide accurate phase-error detection and current-pulse generation.

The Voltage-Controlled Oscillator is modelled to reflect realistic tuning characteristics, phase noise, and gain sensitivity. The Multi-Modulus Divider is implemented to switch dynamically between different integer division factors, while a third order Sigma-Delta Modulator controls this switching to produce an average fractional division ratio. Once all blocks are verified and their metrics such as phase noise and frequency of operation are achieved to be suitable, the complete Fractional-N PLL are to be integrated and simulated in a closed loop with the design of the optimal corresponding Loop Filter. Performance metrics including lock time, frequency resolution, phase noise, jitter, and spur levels are to be analysed using both time-domain and frequency-domain simulation tools. Iterative refinements are made until the design meets the expected specifications.

1.4 Outcome(s)

The obtained outcome of this project is the design and implementation of the fully functional Fractional-N PLL blocks aforementioned, that are capable of producing stable output frequencies with fine resolution and reduced phase noise. The ring based VCO, zero dead zone PFD, wide swing cascode CP, high resolution MMFD and third order Sigma Delta Modulator blocks have all been implemented successfully in the 45nm technology and their outputs have been verified. The integration of a sigma-delta-controlled multi-modulus divider is expected to significantly suppress fractional spurs and shape quantization noise outside the loop bandwidth, along with all the optimizations performed for the other blocks. The PLL will then be expected to demonstrate faster settling time, improved spectral purity, and more flexibility compared to Integer-N counterparts.

1.5 Time Plan

Month	Tasks
January	Literature review on VCO
February	Design of VCO and loop filter
March	Integration of all blocks into the Fractional N-PLL
April	Computing the parameters of the blocks and the PLL
May	Comparison with presently existing architectures and blocks

1.6 Contents of Rest of the Report

The remaining chapters of this report are organized as follows:

- Chapter 1: Provides an introduction to the project domain, explains the objective and subobjectives of the project, provides a timeline for the work done and mentions the results obtained.
- Chapter 2: Provides a state of art analysis with insights from a thorough literature survey from multiple papers.
- Chapter 3: Provides detailed descriptions of design and analysis of each block along with mathematical modelling and design considerations.
- Chapter 4: Discusses the results obtained and analyses the present findings with respect to the performance and design of each block.
- Chapter 5: Provides a final interpretation and conclusion for the work done and discussion of the future scope.

CHAPTER 2: STATE OF THE ART

2.1 Introduction

The current state of the art has many architectures which have evolved from their barebones predecessors to mitigate the issues reported in aforementioned systems, such as fractional spurs and quantization noise from the sigma-delta modulator, phase noise from the VCO, delay from the other sub-blocks, higher power consumptions.

2.2 Literature Survey

The PLL in this work [1] uses a combinational phase detector with transmission gates that resolves the issues of blind zone and dead zone, along with a three-stage current starved voltage-controlled oscillator (VCO).

$$UP = REF \cdot DIV \cdot \overline{L_1} \cdot L_2 + \overline{REF} \cdot \overline{DIV} \cdot L_1 \cdot \overline{L_2} \quad (1)$$

$$DN = REF \cdot DIV \cdot L_1 \cdot \overline{L_2} + \overline{REF} \cdot \overline{DIV} \cdot \overline{L_1} \cdot L_2 \quad (2)$$

In this setup, the charge pump uses a pair of transistors in a current mirror configuration, matching currents to mirror the charge pump current through MOSFET transistors. This mirroring configuration ensures precise and matched currents, promoting a symmetrical charge transfer mechanism. This PLL has been implemented in Cadence Virtuoso.

Very low power dissipation of 8.05mW from a 0.75V power supply, frequency of VCO centered at 2.5GHz, with a phase noise performance of – 127 dBc/Hz at 1 MHz offset. It is intended to achieve phase locking in the sub 6GHz frequency range for use in software defined radio (SDR) and exhibits reduced lock time of ~13.5ns, offering a percentage improvement of 32.5%.

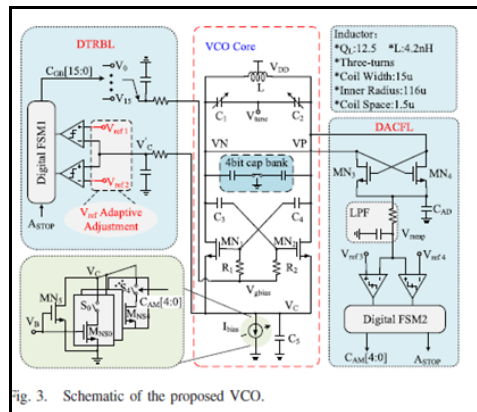


Fig 2.2.1 Schematic of proposed VCO

For the proposed VCO in [2], the authors have improved upon the already existing Class C VCO Structure by using Finite State Machines to achieve control of the amplitude by controlling the bias current, using comparators to reduce effect of the error amplifier on PN. It consists of the VCO Core, the digital amplitude control feedback loop (DACFL) and the digital temperature robust loop (DTRBL). The proposed CP composes of the reference voltage generator, the CP Core, compensating circuits and a switching circuit. This circuit majorly eliminates mismatched currents (which leads to reference spurs) and has larger output impedance due to the presence of AMP_P and AMP_N . A new structure of Phase-Frequency Detector (PDF1) is proposed in [3] to overcome the frequency limitations and the dead zone of the conventional design.

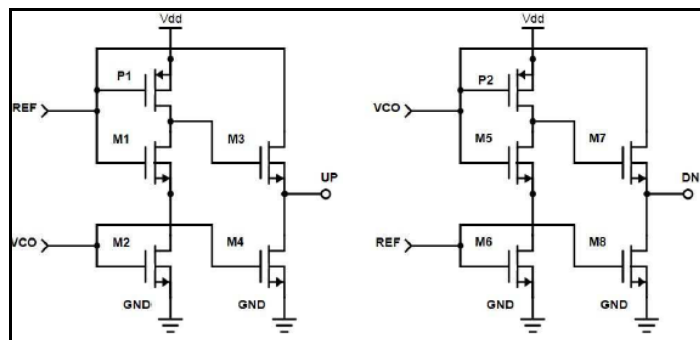


Fig 2.2.2 Proposed PFD Design

The phase noise of the proposed PFD1 which is -133.78 dBc/Hz at 1MHz offset. Compared to conventional PFD (-122dBc/Hz), this proposed PFD1 design

dissipated 8% lower phase noise. 0.18- μm RF CMOS process with an area of $1.07 \times 1.07 \text{ mm}^2$. The measurement results show that the proposed PLL achieves a 2.24–2.85-GHz frequency tuning range and a PN of -123.97 dBc/Hz at 1-MHz offset from a 2.55-GHz carrier. The measured reference spur is -89.4 dBc at 40-MHz offset frequency and the total power consumption is 2.62mW at the 0.8-V power supply voltage. A fast-locking phase-locked loop (PLL) in [4] using the proposed phase error compensator (PEC) is presented. The PEC compensates the accumulated phase error during the frequency acquisition process to enhance the settling time of this PLL. The EPFD is used to track the phase error caused by the frequency error.

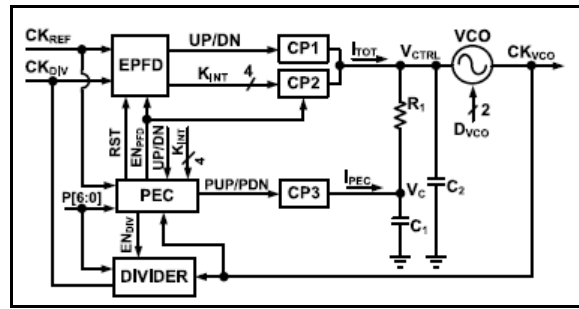


Fig. 2.2.3 PLL using the PEC.

This article [5] presents a fractional-N sampling type-I phase-locked loop (PLL). To overcome the trade-off between reference spur level and PLL bandwidth, the recent trend is to leverage a PLL with a sampling phase detector (SPD). The phase error information is sampled and held on a capacitance and transferred to the loop filter capacitance – implementing a Low Pass Filter.

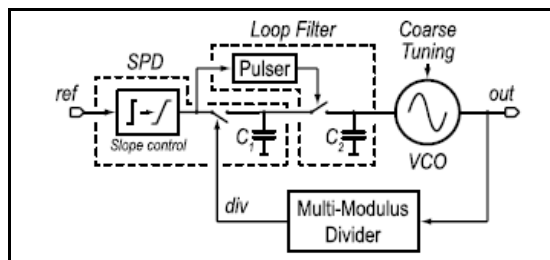


Fig. 2.2.4 Type-I PLL based on SPD.

To overcome the challenge of reliably synthesizing a fractional-N channel an effective way is to realign *ref* and *div* rising edges before phase detection is performed. This is accomplished by cancelling the quantization noise pattern of the $\Delta\Sigma$ modulator used to drive the frequency divider, known *a priori*, via a digital-to-time converter (DTC) circuit. The DTC is digitally calibrated in the background via a sign-LMS algorithm.

The PLL has been fabricated in a 28-nm CMOS technology process, and it synthesizes frequencies from 11.9 to 14.1 GHz, achieving an rms jitter of 58.2 and 51.7 fs (integrated into the 1 kHz–100 MHz bandwidth) for a fractional-N and integer-N channel, respectively. The reference spur is as low as -73.5dBc , while the worst case near-integer fractional spurs are lower than -63.2dBc . With a power consumption of 18mW, the jitter-power figure of merit is -252.1 dB (fractional-N) and -253.3 dB (integer-N). The locking time is below $9\mu\text{s}$ for a 1-GHz frequency step. The synthesizer occupies 0.16 mm^2 , including decoupling capacitors.

An 18.2–18.4 GHz sub-sampling PLL (SSPLL) in [6] is implemented using Taiwan Semiconductor Manufacturing Company Ltd. (TSMC) 90-nm CMOS technology for spin-qubit-based quantum computers. A core loop and a frequency-locked loop (FLL) are used to realize the SSPLL-based frequency synthesizer. The FLL is modified from a classical CPPLL by inserting a dead zone (DZ) creator between the PFD and CP.

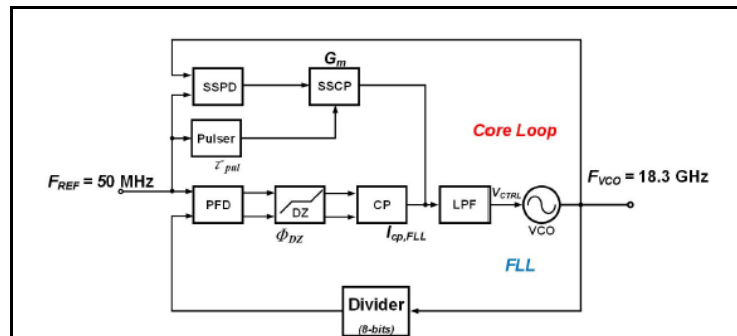


Fig. 2.2.5 Block Diagram of the SSPLL system.

When the SSPLL operates at room temperature, it achieves the in-band phase noise of -96 dBc/Hz at 1 MHz frequency offset and far-out phase noise of -123.5 dBc/Hz at 10 MHz frequency offset for the carrier frequency of 18.3 GHz. By estimation, the frequency noise of 36.1 kHz rms can be achieved at 4 K, which almost meets the requirement of a quantum computer adopting Rabi oscillation of 10 MHz and pulsed microwave signal with a pulse duration of 50 ns for the gate operation.

A frequency-tunable phase-locked loop (PLL) in [7] applied in high-energy particle detectors is developed and explained. To achieve high detection efficiency, high requirements are promoted for an on-chip clock in CMOS pixel sensors, which requires high precision and low power consumption for the PLL. The proposed PLL employs a fractional-N frequency division structure. An LC voltage-controlled oscillator (LC-VCO) is used to generate the local clock. A third-order MASH sigma-delta modulator is used. In the prototype chip, a low dropout regulator (LDO) is integrated to suppress the power supply noise.

The LDO architecture is composed of a bandgap reference, an error amplifier, an output NMOS transistor, and a load capacitor. A feedback mechanism has been adopted to regulate the generated supply voltage. With respect to the LDO, the power source rejection ratio (PSRR) and stability are the two most important indicators. To improve the PSRR by using the NMOS, an error amplifier with increased loop gain and power rejection ratio is used. This can relax the design requirement of the bandgap. In the proposed design, a folded cascode amplifier with a high open loop gain is adopted as the error amplifier.

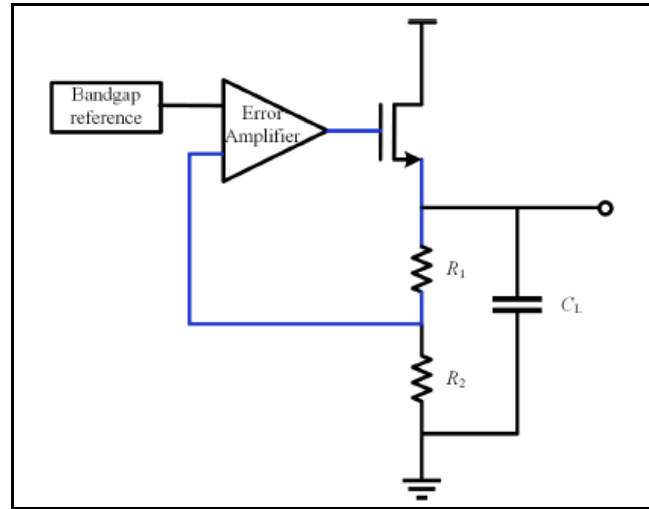


Fig. 2.2.6 LDO Circuit Architecture

The PLL was designed and fabricated in a 180-nm CMOS process. It costs an area of 1.29 mm² and a power of 8.1 mW at a 1.8-V power supply. The measured output frequency range is 400–530 MHz with a reference frequency of 25–35 MHz. The rms jitter, integrated from 100 kHz to 100 MHz, is 4.8 ps at 526 MHz. The in-band spur is less than –61 dBc, and the PN at 1-MHz frequency offset is less than –113 dBc/Hz. This paper [8] discusses the design and simulation of a Delta-Sigma based fractional N PLL frequency synthesizer for 2.4 GHz wireless standards. The frequency response of open and closed loop gain in fractional N-PLL circuits is simulated using a MATLAB application. A single-loop Sigma-Delta modulator of second order Using a 4-level quantizer, the proper fractional ratio (α) was obtained.

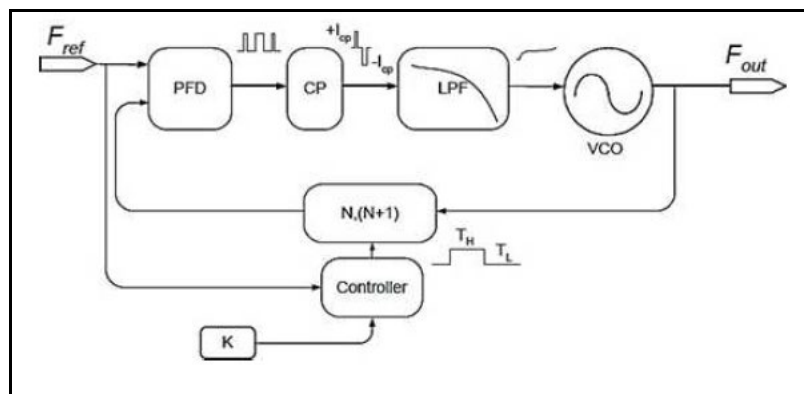


Fig. 2.2.7 Architecture of PLL

The development of the loop filter of second order using the following components: $C1 = 1263.9 \text{ pF}$, $C2 = 6881.4 \text{ pF}$, and $R1 = 0.5871 \text{ K}\Omega$. The values were obtained using the following variables: $\text{pm} = 47^\circ$, $K\phi = 100 \text{ }\mu\text{A}$, $K_{\text{vco}} = 760 \text{ MHz}$, $F_{\text{ref}} = 40 \text{ MHz}$, and $\text{BW} = 100 \text{ kHz}$.

This paper [9] presents a frequency synthesizer for generating quadrature local-oscillator (LO) waveforms covering an octave range of 6 to 12-GHz. It uses two voltage-controlled oscillators (VCO) to cover an octave range from 12 to 24 GHz. The VCOs employ inductor mode and capacitor switching to obtain a wide tuning range while maintaining a good phase noise FOM. A digital technique is introduced to counter modulus-dependent delay (MDD) in the feedback divider, which can exceed 100 ps. MDD is basically the delay introduced by each 2/3 divider cells in the feedback divider.

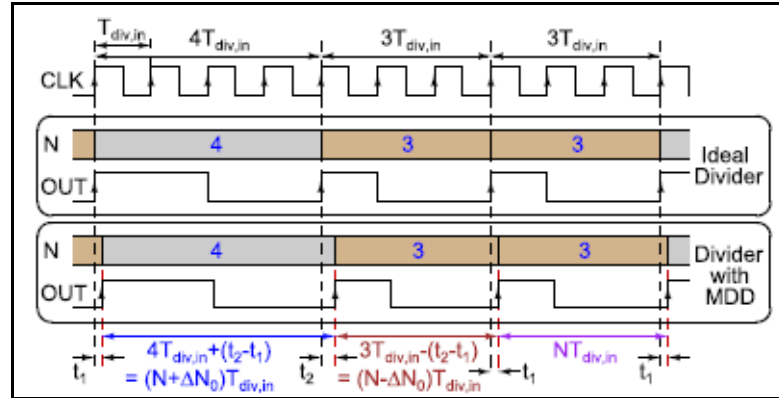


Fig. 2.2.8 Timing diagram

MDD leads to increased jitter and phase noise due to dynamic changes in divider delay in [9]. Existing solutions include resynchronizing the divider output and using feedback signals, but they still exhibit residual MDD. The proposed method involves pre-distorting the fractional input to the digital delta-sigma modulator (DDSM) to counteract MDD. The technique transforms MDD into a periodic error at half the reference frequency, which can be filtered out effectively. The following comparison is made:

COMPARISON WITH VCOs IN THE RECENT LITERATURE							
References	This work 12–17GHz VCO	[20] JSSC '23	[21] JSSC '23	[22] JSSC '23	[23] TMITT '23	[24] JSSC '22	[13] ISSCC '19
Technology	65 nm CMOS	65 nm CMOS	28 nm CMOS	65 nm CMOS	55 nm CMOS	28 nm CMOS	65 nm CMOS
Power (mW)	17.07	17.25	46	29.2	22.5–23.6	173	17.5
Frequency (GHz)	11	17.7	17.6–19.4	25.7	30.7	52.4–60.4	10.65–14.37
Tuning range (%)	46.7	9.7	18.1	14.2	29.7	27.4	41.2
PN ^a at 1MHz (dBc/Hz)	-112.7	-109.4	-116.6	-119.7	-118.9	-120.2	-115.4
FoM [#] (dB)	180.4	177	181	185	184.3	182.2–186.5	179–182.9
FoMT ^b (dB)	193.8	190.4	180.8	190.2	189.4	185.2–189.5	188.3–192.2
						193	199.3

^a Normalized to carrier frequency of 10 GHz; [#] FoM = $|\text{PN}(\text{dBc}/\text{Hz})| + 20\log_{10}(f_0/\Delta f) - 10\log_{10}(\text{Power}(\text{mW}))$;

Fig. 2.2.9 Comparison Table

This paper [10] presents a novel phase-locked loop (PLL) design that effectively reduces jitter and reference spur using a fast phase-error correction technique. The proposed fast phase-error correction (FPEC) technique emulates the phase-realignment mechanism of injection-locked clock multipliers (ILCMs) to remove accumulated jitter quickly. Injection-locked clock multipliers (ILCMs) can achieve low jitter but are sensitive to frequency drifts, leading to increased reference spur.

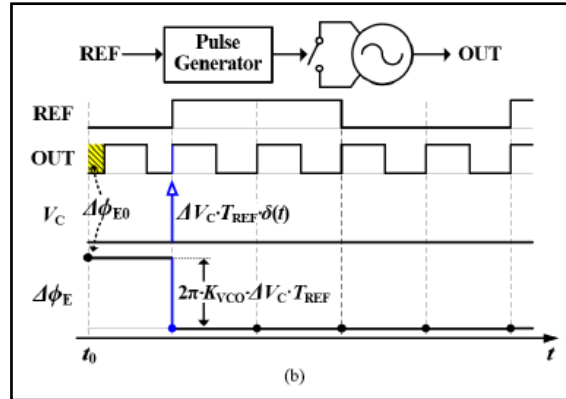


Fig. 2.2.10 Phase Error Timing Diagram

This paper [11] discusses the design and implementation of an ultra-low-voltage phase-locked loop (ULVPLL) that operates at supply voltages as low as 0.25 V, aimed at enhancing power efficiency for IoT applications. Prior ULVPLLs experience significant performance drops when VDD is below 0.3 V. All-digital ULVPLLs suffer from poor time-to-digital converter (TDC) resolution at low voltages. Analog ULVPLLs face voltage headroom limitations and current mismatch issues at sub-0.3 V. The trade-off between loop bandwidth and spur

level remains a challenge in existing designs. The architecture consists of a proportional path (P-path) for phase tracking and an integral path (I-path) for frequency acquisition. The design allows for a wide loop bandwidth to effectively suppress phase noise without degrading spur levels. The P-DCP operates without current sources or mirrors, reducing power consumption and complexity.

This paper [12] presents a low power PFD implemented in 90nm technology. Traditional PFDs utilize D flip-flops and AND gates, leading to drawbacks such as non-zero dead zones and high-power consumption at elevated frequencies. The proposed PFD design eliminates the reset path, directly connecting input signals to D flip-flops, which enhances performance by reducing reset time and dead zone. Operates effectively at frequencies ranging from 10MHz to 4GHz, with a total power consumption of 10.85 μ W at 0.1GHz and a supply voltage of 1V.

This paper [13] discusses about a PLL achieving an output frequency of 7.2GHz and a power consumption of 3.09mW. It is implemented in Cadence Virtuoso using SCL 180nm technology. A single-ended ring VCO and was chosen for small chip size, low power consumption and wide frequency range. The core blocks are as follows: Phase Frequency Detector (PFD), Charge Pump (CP), Low Pass Filter (LPF), VCO, Frequency Divider. The divider is built with D-flip flops and divides 6.4GHz VCO output down to 200MHz reference.

This paper [14] provides a comprehensive review of PLL architectures and classifies them based on topology and functional criteria. PLL basically synchronizes output frequency with a reference signal using a feedback loop.

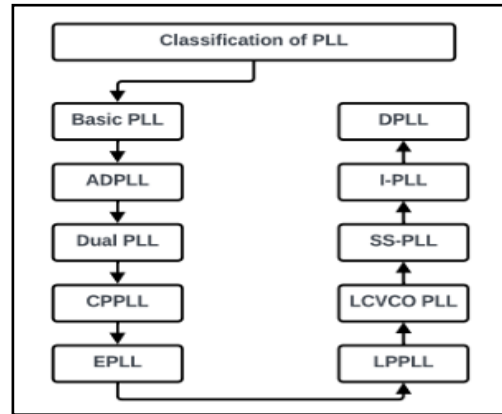


Fig. 2.2.11 PLL Classifications

This paper [15] presents a current starved sleep voltage-controlled oscillator (VCO) for the Phase Locked Loop (PLL) at high frequency with low power. The proposed VCO uses modified delay cells—namely stack delay cell, sleepy stack delay cell, and sleep delay cell structures—to significantly reduce leakage power consumption while maintaining a wide tuning range and stable oscillation frequency. The study concludes that integrating sleep transistor techniques into VCOs provides a viable method for energy-efficient and high-performance clock generation in modern mixed-signal and RF systems. The design exhibits improved phase noise performance of -115 dBc/Hz at 1 MHz offset and a figure of merit (FoM) of -228.2 dBc/Hz, indicating high efficiency and stability for PLL applications.

This paper [16] presents a design of a hybrid ring-type CSVCO. Two VCO designs are proposed. First is a current starved pull-up sleepy ring VCO which offers improved frequency of oscillation and reduced phase noise. The second is a hybrid current starved ring VCO with the bulk-driven keeper technique which yields better results by reducing the power consumption and mitigating leakage current. The results clearly demonstrate that the proposed design is extremely tolerant to PVT (process, voltage, and temperature) variations making it well suitable for high-frequency PLL applications.

This paper [17] presented a dual mode Multi-Modulus Divider (MMD) in 65nm technology consisting of divide-by-2/3 cells cascaded in a frequency synthesizer that can support maximum operation up to 11.4 GHz. Unlike in traditional divide-by-2/3 cells, these cells used NOR gates instead of NAND gates to avoid stacking of differential pairs, improving common-mode compatibility. The fractional functionality was realized through a DSM. The MMD provides a division ratio range from 128 to 255 and 512 to 1023, achieving a maximum working frequency of 11.4 GHz.

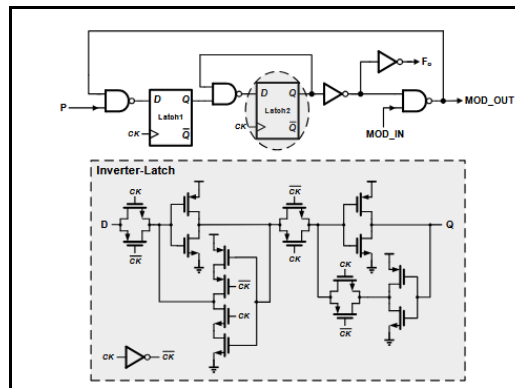


Fig. 2.2.12 Inverted based 2/3 cell schematic

Phase frequency detectors (PFDs) are essential components of phase-locked loop (PLL) systems. PFDs create outputs depending on the phase difference between their two input signals after comparing the waveforms [18]. The voltage-controlled oscillator (VCO) output signal and the reference signal are used as input signals in a PFD, while the UP and DOWN signals are used as output signals. Two PFD are projected that show how dead zone alters working of frequency synthesizer. A conventional charge pump with switched current sources is used to drive the capacitive load. Simulation results are provided for both typical and extreme conditions (temperature and voltage variations), demonstrating reliable circuit operation. Addition of additional electrical converter delays within the reset path helped avoid the potential drawback of a “dead zone” within PFD response. The two PFD also succeeds in lower power consumption.

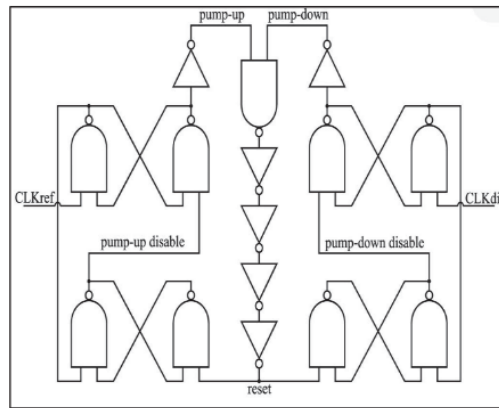


Fig. 2. 2.13 NAND bases PFD

[19] Developed an enhanced design for a Current-Starved Ring VCO that operated efficiently in a high-frequency range using 45nm CMOS technology. The primary focus was to achieve minimal phase noise, a broad tuning range, and enhanced power efficiency. The study reached to the conclusion that while maintaining the length of 45nm and varying the width across 0.13 μ m, 0.2 μ m and 0.4 μ m, all the observed parameters show better performance. This was attributed the transistor having decreased thermal noise and gate capacitance with smaller widths. The 0.13 μ m VCO demonstrated superior phase noise performance at -81.7dBc/Hz, suggesting reduced noise and improved signal quality. The phase noise of the 0.4 μ m VCO was found to be -75.7dBc/Hz, which can be attributed to factors such as increased thermal noise and gate capacitance. The wider the transistor, the higher the power consumption. The VCO with a size of 0.4 μ m consumed the highest power at 149.5 μ W, whereas the VCO with a size of 0.13 μ m consumed the lowest power at 120 μ W.

This paper [20] proposes an architecture for an improved voltage controlled oscillator (VCO) whilst comparing the performance of a conventional ring VCO in 130nm technology. The conventional ring oscillator consists of inverter as delay stages and number of delay stages is connected with each other in the form of a chain. The output of last stage is fed back to the input of first one. The novelty is in the design proposed where there is an addition of an extra nMOS transistor based ring VCO. The variable resistance is implemented using nMOS transistor and added at the input terminal of each inverter.

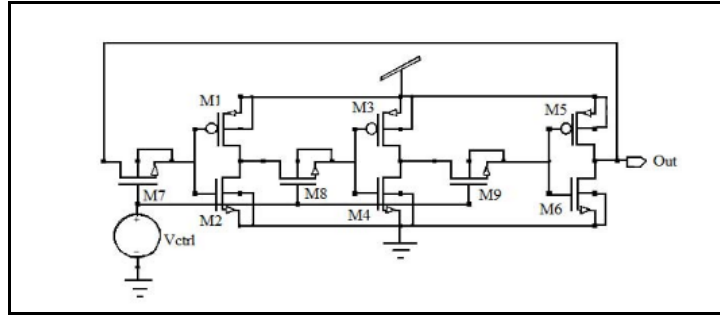


Fig. 2.2.14 Proposed three stage ring VCO

The conclusion was found to be that the proposed model performed better compared to the existing model, exhibiting output frequency from 2260.65MHz to 3441.71 MHz with variation in power consumption from 31.08 μ W to 52.21 μ W, compared to 338.98 MHz to 1208.73 MHz from the latter.

An inductorless Harmonic Mixer (HM) based phase locked loop was proposed in this paper [21]. Inductorless PLLs are appealing due to their smaller area requirements and wider tuning range, yet they are difficult to design for Fractional-N PLLs due to conflicting bandwidth requirements for suppression of DSM quantization noise and VCO phase noise. To overcome the insufficient DSM noise suppression in HM-based PLLs and allow a wider loop bandwidth for the suppression of ring VCO phase noise, the architecture is such that feedback divider was split into two parts and insert a nested PLL in between. In the proposed architecture, the feedback gain of the main PLL is dominated by the HM based path, meaning that the nested PLL has a negligible effect on the overall open-loop transfer function and stability, and its bandwidth can be made narrow to effectively filter out the DSM noise. A 2.8-3.5GHz fractional-N synthesizer based on the proposed architecture was fabricated in 65-nm CMOS as a proof-of-concept, which achieved a -227.6dB FoM with an 8 MHz bandwidth, with no calibration circuitry and with a compact layout containing no inductors.

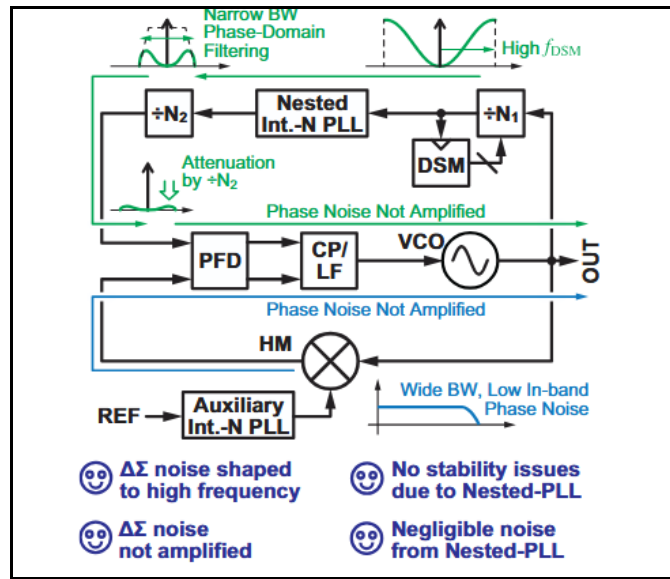


Fig. 2.2.15 Proposed architecture with its noise suppression

The PLL architecture proposed in this paper [22] consists of cascaded PLLs with Ring Oscillator based DLL and type-II PLL as first and second stages, where the first stage served as a frequency multiplier and the second stage consists of Delta Sigma Modulator (DSM). The increase in operating frequency of the DSM suppresses its quantization noise. The Phase Noise (PN) of the first stage is improved to achieve a PN multiplication factor removal using a Burst Mode Sampling scheme, also relieving PFD/CP noise factor of the main fractional-N. This was all implemented in 28nm CMOS technology and the PLL prototype occupies a 0.016 mm² active area, achieving a 686 fs integrated rms jitter from 10KHz to 40MHz at a 4 GHz output frequency; while consuming 10.21mW with -233.6 dB FoM jitter. The measured fractional and reference spurs are -59.8 dBc and -54.5 dBc, respectively.

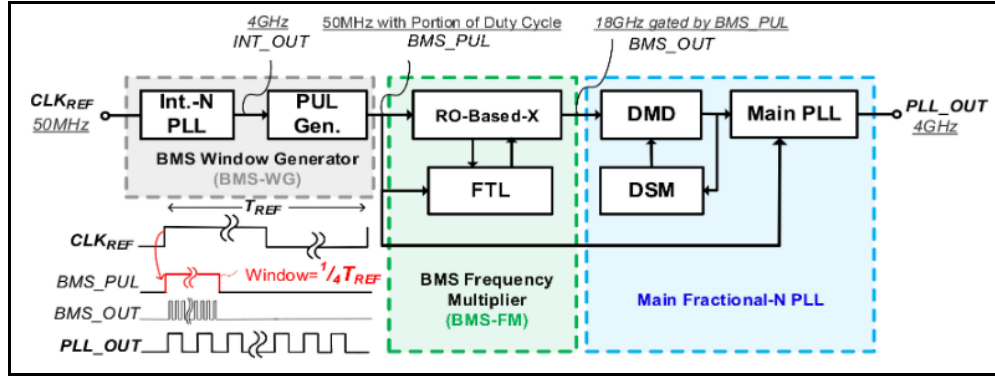


Fig. 2.2.16 Top architectures and timing diagrams of BMS Fractional-N PLL

This paper [23] proposed a new fractional-N frequency synthesizer for 5G+ communication systems, where unlike a classical RF synthesizer, the proposed synthesizer frequency multiplication is performed by an intermediate frequency (IF) voltage-controlled oscillator (VCO) and a programmable-K frequency multiplier (FM). The design tackles the industrial demand for high-speed wireless communication by tackling present issues in three parts- phase synchronization, frequency multiplication and frequency division. The presented synthesizer was fabricated on a 130-nm SiGe process, and its core die size was 1 mm². The total power consumption is 15mW, as well as the channel resolution was 390.625 kHz to guarantee less than -75dBc fractional spur. The integrated rms jitter for the 1 kHz-100-MHz offset frequency band was measured as 77 fs.

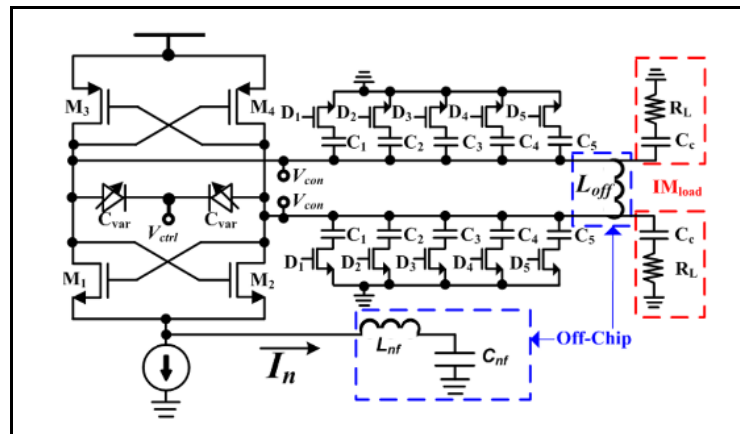


Fig. 2.2.17 IFVCO Schematic

The proposed PLL [24] employs a fractional-N frequency division structure. An LC voltage-controlled oscillator (LC-VCO) was used to generate the local clock. A third-order MASH sigma-delta modulator (6-1) was used to alleviate the quantization noise introduced by the fractional-N frequency divider. To further reduce the phase noise (PN), preset and post set binary frequency dividers were employed. A frequency-tunable phase-locked loop (PLL) applied in high-energy particle detectors has been developed. To achieve high detection efficiency, high requirements are promoted for an on-chip clock in CMOS pixel sensors, which require high precision and low power consumption for the PLL. In CMOS pixel sensors, stable and precise clock signals are crucial for precise time measurement and fast readout. Due to its frequency multiplication capability, the phase-locked loop (PLL) becomes the best choice as an on-chip clock generator for the CMOS pixel sensors. The PLL was designed and fabricated in a 180-nm CMOS process. It costs an area of 1.29 mm² and a power of 8.1mW at a 1.8-V power supply. The measured output frequency range is 400–530 MHz with a reference frequency of 25–35 MHz. The rms jitter, integrated from 100 kHz to 100 MHz, is 4.8ps at 526MHz. The in-band spur is less than –61dBc, and the PN at 1-MHz frequency offset is less than –113dBc/Hz.

This paper [25] brought to light an all-digital fractional-N bang-bang PLL (BBPLL) that enabled low-complexity and low-power frequency generation and modulation. By integrating the FIR filtering in the ILO-based PDLPF, the quantization noise effect of the 16 modulator was significantly mitigated in the fractional-N BBPLL, resulting in good in-band phase noise and fractional spur performance. The proposed method is also useful to reduce the out-of-band phase noise for conventional fractional-N PLLs. A prototype fractional-N BBPLL was implemented in 65-nm CMOS. The in-band phase noise and the fractional spur were improved by more than 25 and 10 dB, respectively. An in-band noise of –96.6dBc/Hz and a reference spur of –78.7dBc were achieved at 2.6-GHz output, consuming 4.3mW from a 0.9-V supply.

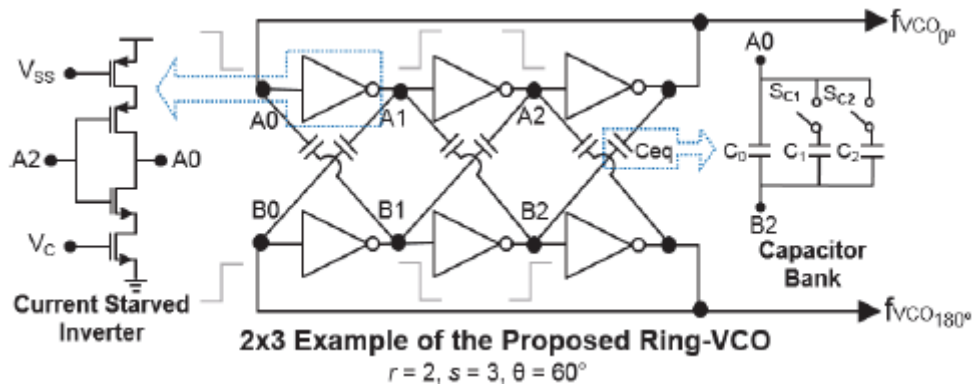


Fig. 2.2.18 Interconnected coupling capacitor ring VCO

This paper [27] implements the ring VCO with 3 main building blocks. They are as follows, current-starved inverters for control voltage tuning, two or more rings each made up of an odd number of current-starved inverter stages, and small symmetrically laid-out interconnect capacitors that couple the phases of neighboring input and output nodes of the rings. The key design improvement is the addition of coupling capacitors placed between ring stages which form a continuous virtual inductor through a distributed spiral arrangement. This causes the energy lost during each cycle to be restored at the phase angle that adds minimal noise, which is the exact opposite of a conventional ring oscillator where energy is added at the most jitter-sensitive phase angle. This is what separates it from a standard ring oscillator, it behaves more like an LC oscillator without using any physical inductors. It also offers several advantages over conventional ring oscillators: precise quadrature output with many additional phase outputs available, wide tuning range, inductor-like quality and stability without using physical inductors, full scalability to deep sub-micron IC process nodes, compact area with minimal-sized inverters, and operation at supply voltages of 1V and below with extremely low power consumption. The proposed 4x3 quadrature ring VCO achieves a frequency range of 0.8 to 28.2 GHz, occupies an area of 0.0024 mm², consumes 0.88 mW at a 1.0V supply, and achieves a phase noise of -124.5 dBc/Hz at 10 MHz offset for a carrier frequency of 28.0 GHz.

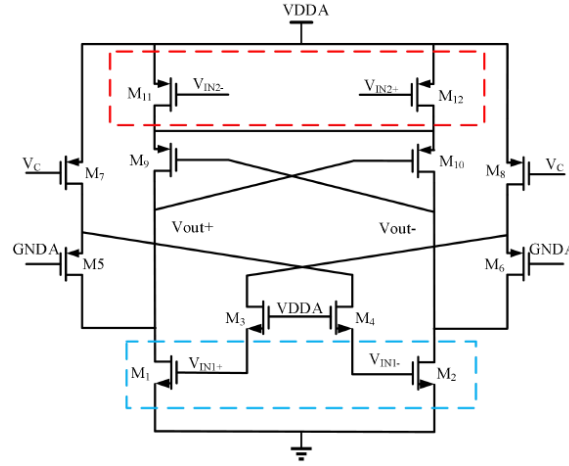


Fig. 2.2.19 Pseudo-differential delay cell

This paper [28] proposes an architecture that replaces the traditional single feedback loop with a dual feedback structure consisting of a primary feedback loop and an auxiliary feedforward path. The delay cell uses pseudo-differential input pairs instead of a conventional tail current source, which reduces fixed static power consumption. Cross-coupled transistors create a positive feedback latch that accelerates waveform establishment and minimizes noise current injection during signal transitions — the primary contributor to $1/f^3$ phase noise. The VCO is simulated in TSMC 180nm CMOS with a linear tuning voltage range of 0 to 1.2V, an oscillation frequency spanning 0.9 GHz to 2.65 GHz, a phase noise of -93.1 dBc/Hz at 1 MHz offset for a 2 GHz center frequency, and a power consumption of approximately 1.89 mW, yielding a FoM of 156.7 dBc/Hz. The total layout area excluding the output buffer is $80 \mu\text{m} \times 45 \mu\text{m}$.

2.3 Summary of the Literature Survey

This literature survey details recent advancements in Phase-Locked Loop (PLL) design, focusing on innovations to overcome core challenges like phase noise, jitter, power consumption, and fractional spurs. Key improvements are seen at the sub-block level, including novel Phase-Frequency Detectors (PFDs) that eliminate dead zones, charge pumps with enhanced current matching to reduce reference spurs, and a variety of Voltage-Controlled Oscillators (VCOs)—such as current-

starved, Class-C, and ring-based architectures—aimed at achieving wider tuning ranges and lower phase noise.

At the architectural level, the survey highlights a strong emphasis on fractional-N PLLs using sigma-delta modulators for fine frequency resolution. To mitigate the inherent quantization noise of these systems, advanced techniques like Digital-to-Time Converters (DTCs) for noise cancellation and sub-sampling phase detectors (SSPDs) for superior noise performance are employed. Furthermore, multi-stage and nested PLL architectures are explored to effectively suppress different noise types, allowing for wider loop bandwidths. The designs are highly application-specific, targeting areas from 5G communication and software-defined radio to quantum computing and ultra-low-power IoT devices, with reported performance metrics often achieving jitter below 60 femtoseconds and power consumption as low as 2.62mW. The aim is to use optimized architectures that fit the parameters we are looking for and improve upon their design in our project while overcoming common challenges seen in this survey.

CHAPTER 3: DESIGN AND ANALYSIS

3.1 Introduction

This chapter includes the design blocks of the different blocks used in the PLL - the Phase Frequency Detector, Charge Pump, VCO, MMFD and the Sigma Delta Modulator. It provides comprehensive analysis of the blocks with appropriate expressions and formulae. Conceptual block diagrams, gate level diagrams and switch level diagrams have been integrated to give detailed information about the system.

3.2 Phase Frequency Detector and Charge Pump

Phase error detection and correction is an important part of the phase locked loop system. The Phase Frequency Detector (PFD) performs this error detection by comparing the reference frequency with the output frequency generated by the VCO. Based on the error detected, the PFD generates up (UP) and down (DN) signals, driving the Charge Pump (CP) which sources or sinks the current into/from Loop Filter (LF).

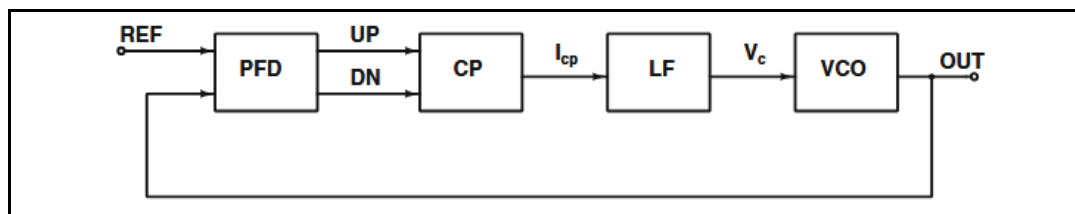


Fig. 3.2.1 PLL

The LF then filters out high-frequency components and generates a control voltage which is proportional to the phase error between the reference and feedback signal. The voltage drives the Voltage Controlled Oscillator (VCO) output frequency in the direction of reducing the phase error.

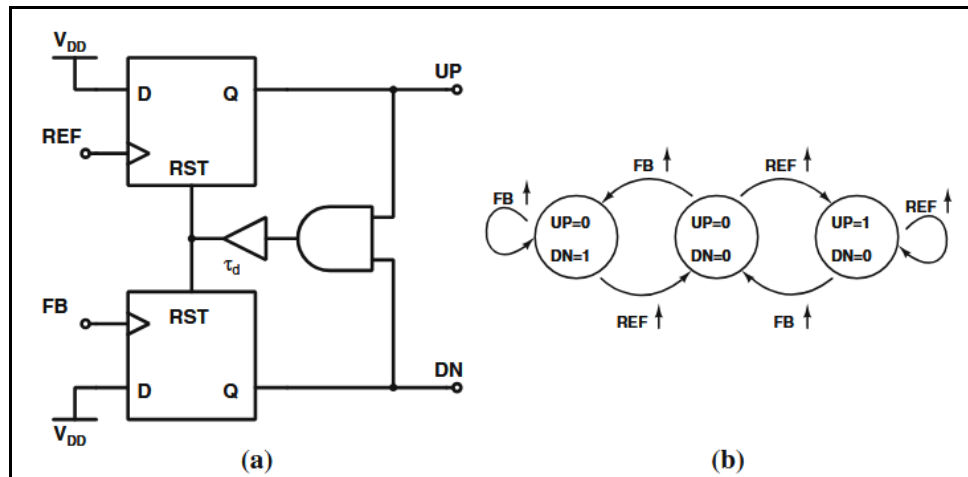


Fig. 3.2.2 Basic architecture of the conventional PFD-CP configuration.

The PFD encounters a dead zone when there is a delay in the reset edge of the generated UP and DN signals, usually due to propagation delay in the PFD circuit. Dead zone is undesirable because it leads to improper locking of the PLL, generating a slightly oscillating voltage at the output instead of a DC voltage which leads to phase noise. This dead zone is avoided by biasing the opposite of the generated pulse for a short duration such that both signals are on and lead to source and sink currents being equal for that period of time where clear edge is supposed to fall. Thus, since both currents will be equal, the voltage generated will be stable. The circuit implemented in this project is a zero dead zone PFD. Push-Pull Complementary Logic (PPCL) gates have been used since they have low rise and fall times, curbing the propagation delay that we may face and help in eliminating the dead zone.

The CP that is cascaded with the PFD uses wide-swing cascode current mirrors for sourcing and sinking currents for higher output impedance and to reduce current mismatch.

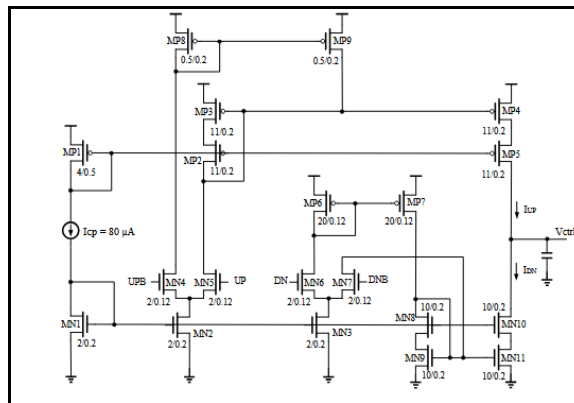


Fig. 3.2.3 NMOS Wide Swing Cascode Charge Pump

Alongside this modification, the switching circuit was constructed using transmission gates to reduce propagation delay. These modifications were made in the circuit to improve its overall speed and performance to give a stable current to the LF.

The following schematics were designed in Cadence Virtuoso:

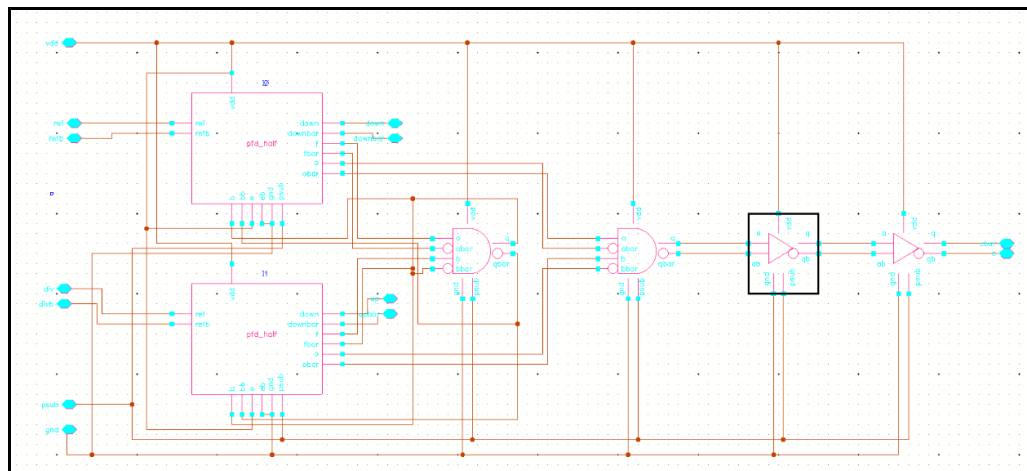


Fig. 3.2.4 Designed Phase Frequency Detector

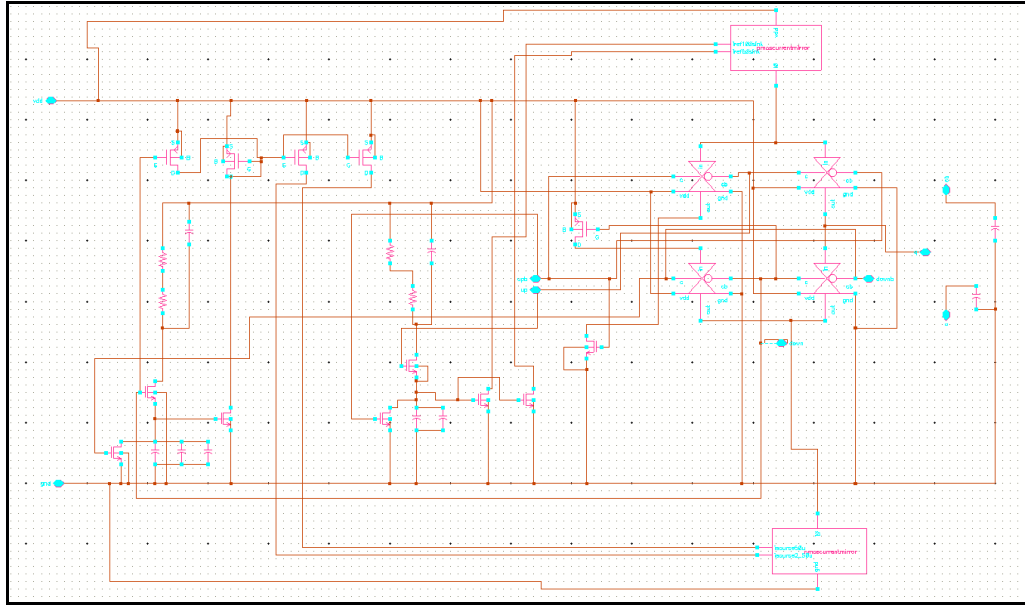


Fig. 3.2.5 Schematic of the designed Wide Swing Cascode Charge Pump

3.3 Voltage Controlled Oscillator

Voltage Controlled Oscillators (VCO) are the critical building block in high-speed systems. A ring-based architecture is widely used due to its simple design and ability to offer a wide tuning range. The ring oscillators are also more compatible with CMOS technologies since they consume lesser silicon area compared to traditional LC oscillators. The VCO generates a frequency based on the input voltage. The VCO frequency changes as a function of the supply voltage, process and temperature.

A ring oscillator consists of a series of delay stages connected in a closed loop fashion such that the signal continuously circulates, producing periodic oscillations. For sustained oscillations, the Barkhausen criteria must be satisfied which is as follows,

$$|H(j\omega)| \geq 1$$

$$\angle H(j\omega) = 180^\circ$$

The oscillation frequency of a N-stage ring VCO is determined by the cumulative propagation delay of its delay stages. For an N-stage ring oscillator, the frequency is given by the following formula:

$$f_{osc} = \frac{1}{2 N \tau_d}$$

The delay cells used in ROs can be classified into 2 types:

1. Single-ended inverter-based delay cell
2. Differential delay cells

While the single-ended is pretty straightforward and occupies lesser area, it is highly sensitive to noise on the power supply and substrate, since the output of each stage is referenced to either VDD or GND directly. Any noise on these lines, directly corrupts the oscillation which leads to poor phase noise. A differential delay cell on the other hand processes the signal as a difference between 2 complementary nodes (V+ and V-). Since both the nodes experience the same common mode noise, these cancel out at the differential output, thus giving them a superior common mode noise rejection, better power supply noise immunity and the ability to naturally produce quadrature outputs from the ring.

The implemented differential delay cell is shown below in Fig. 3.3.1 (a). It is a pseudo-differential structure that is free of the conventional tail current source, that is responsible for limiting the output voltage swing and the 1/f noise sensitivity. The cell operates through 2 feedback paths, M1 and M2 form the primary differential input, suppressing common mode noise from the power supply and substrate injection. M11 and M12 serve as auxiliary inputs to the ring oscillator, controlling transistors M9 and M10 to create a positive feedback latch circuit, reducing the overall static current. The control voltage (VC) varies the delay cell's current by controlling transistors M7 and M8, thereby altering the delay time of the delay element. Additionally, within this delay cell, M1, M3, M6, and M2 form another current-extraction positive feedback system. This system can expedite the establishment of the waveform, reduce noise current injection during signal conversion, and thereby improve phase noise. [28]

The 4 differential delay cells are cascaded in a loop manner to form the ring VCO as shown in Fig. 3.3.1 (b).

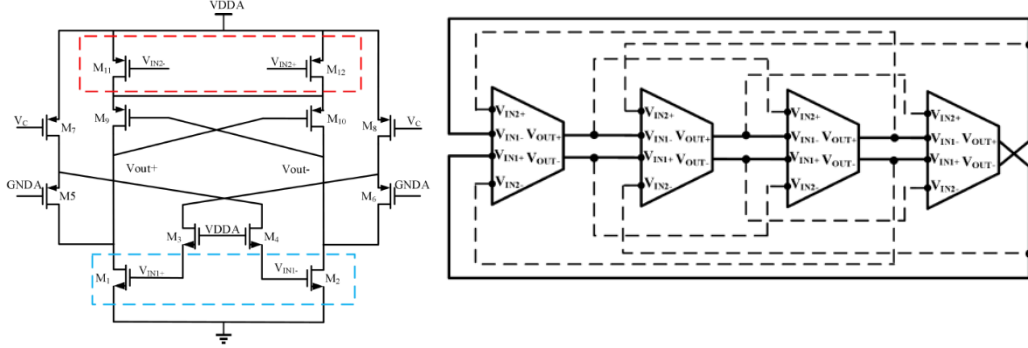


Fig. 3.3.1 (a) Circuit diagram of differential delay cell & (b) 4-stage ring VCO using differential delay cell

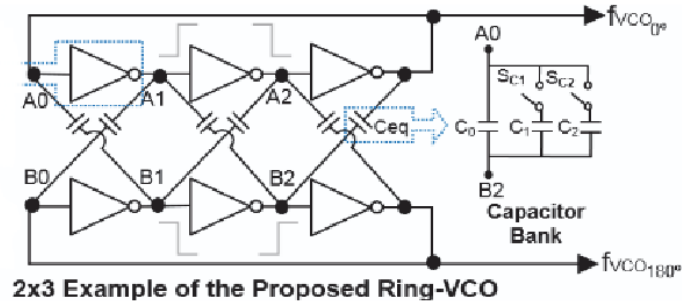


Fig. 3.3.2 Ring VCO architecture with interconnected coupling capacitors

The ring VCO mentioned in [27] uses phase injection locking via a network of symmetrically placed passive metal interconnect coupling capacitors to reduce the phase noise of an inverter-based ring VCO (Fig. 3.3.2). The result of the proposed RO design is a more energy efficient circuit which evenly distributes charge between the various nodes during oscillation. The phase injection-locking via capacitive coupling to provides the VCO with improved phase noise performance and spectral purity properties than state of the art RO designs, making the proposed ring VCO design more comparable to those of LC-based ones. A capacitor in the inter-row feedback path behaves inductively due to the gyrator effect, creating a distributed virtual inductor across the ring. This causes the VCO

to operate in a more sinusoidal, linear mode similar to an LC oscillator where energy lost per cycle is restored at the phase angle of minimum jitter, rather than at the most noise-sensitive instant as in a conventional ring oscillator. This brings the phase noise performance of the ring VCO significantly closer to that of LC-based designs, without requiring any physical inductors. The below circuit is the implemented 4-stage, ring VCO with interconnected coupling capacitors with differential delay cells.

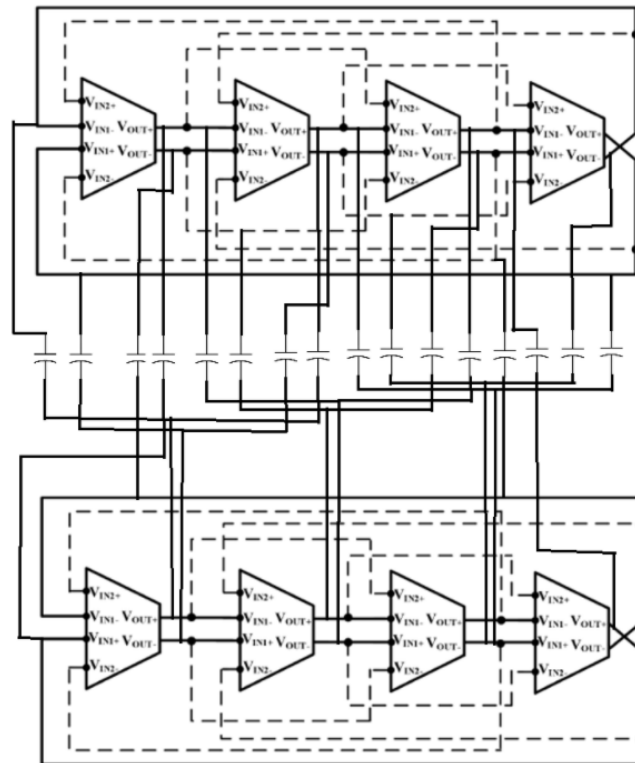


Fig. 3.3.3 Circuit diagram of implemented ring VCO

3.4 MMFD

A Multi-Modulus Frequency Divider (MMFD) is a key block in phase-locked loops (PLLs), frequency synthesizers, and clock/data recovery systems. Unlike fixed-ratio dividers, an MMFD allows the division ratio to be programmable using digital control signals. Controlled by the sigma-delta modulator, the MMFD rapidly alternates its division value so that the average division ratio matches the desired fractional value [17]. This approach allows the PLL to achieve frequency steps much smaller than the reference frequency while maintaining a high loop bandwidth. Proper MMFD design is crucial because timing mismatches, glitches, or unequal delay between division paths can generate fractional spurs and degrade phase noise. In a PLL synthesiser, the MMFD is placed in the feedback path (or in the reference path) to divide the VCO (Voltage Controlled Oscillator) output by N . This programmability enables:

- wide tuning ranges
- fine frequency steps
- multi-band operation
- reduced lock time in PLLs

MMFD is designed using a simple pass transistor logic, to save on space used by high-speed logic such as CML and TSPC and also to cut power usage. The most widely used implementation is the $2/3$ divider cell, which forms the building block of cascaded MMFD chains. It is the fundamental block of an MMFD, which can operate in:

- Divide-by-2 mode
- Divide-by-3 mode (activated once per cycle by a control signal)

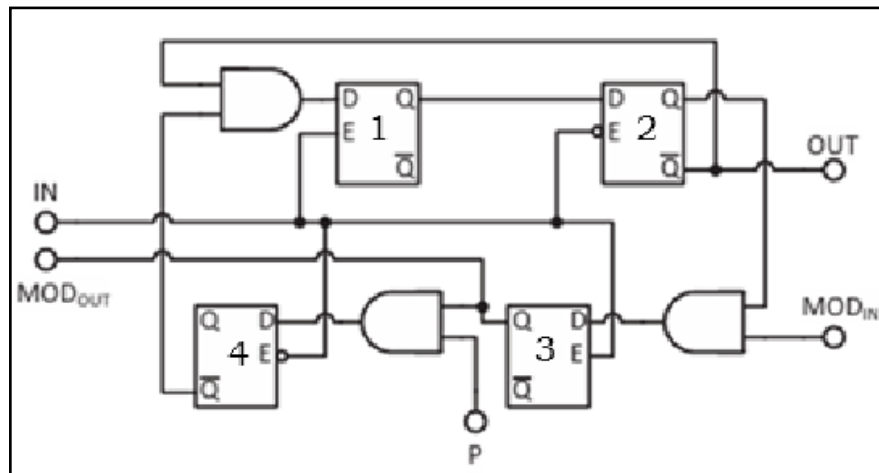


Fig. 3.4.1 Block diagram of 2/3 divider cell

A master-slave configuration of D-Flip Flop is implemented using D-Latches to obtain the division of the input clock signal. P is the control bit that enables division by either 2 or 3 in the 2/3 divider cell. When P or MODin bits are set to logic 0, the input clock signal gets divided by 2. When P and MODin are set to logic 1, a division by 3 occurs.

The 2/3 divider cell is made of:

- D-Latches designed using Transmission gates.

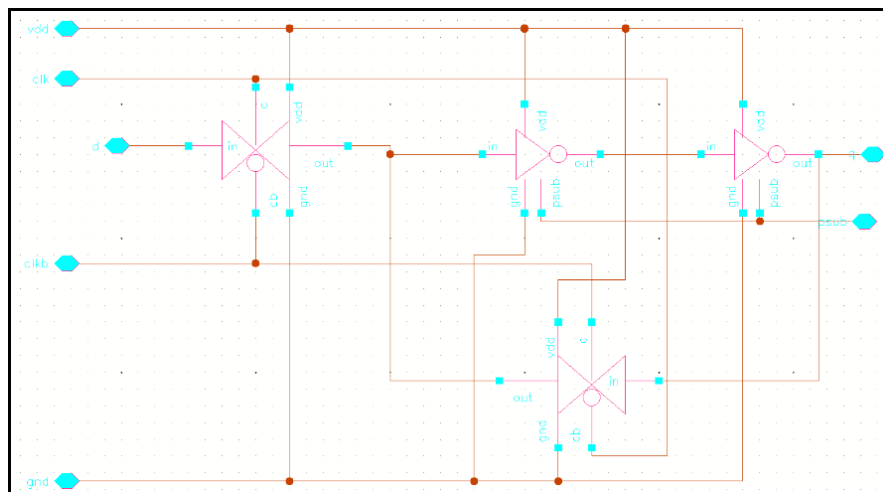


Fig. 3.4.2 Schematic of designed D-Latch in Cadence

Virtuoso

- NOT gates
- AND gates designed using NAND and NOT

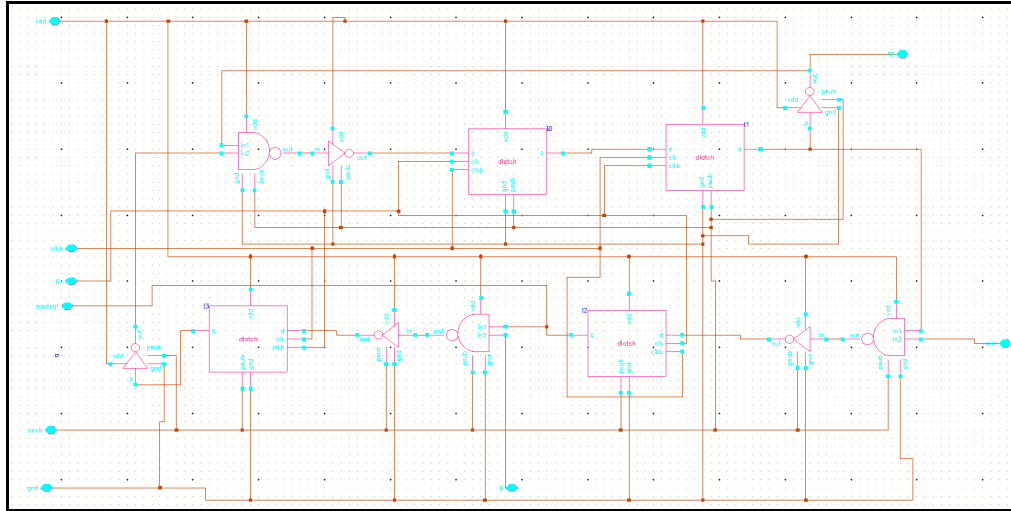


Fig.3.4.3 Design of 2/3 divider cell in Cadence Virtuoso

By connecting the 2/3 cells in a cascaded manner, required divide ratio is achieved. Where: $P_0, P_1, P_2 \dots$ are the modulus control bits and each control bit decides whether that stage divides by 2 or 3 at a specific moment. As the control

$$DivideRatio = 2^n + 2^{n-1}P_{n-1} + 2^{n-2}P_{n-2} + \dots + 2^1P_1 + 2^0P_0$$

bits (P_0, P_1 etc.) change, the divide ratio changes, thereby changing the output frequency of the PLL. This allows fine frequency steps and rapid band switching.

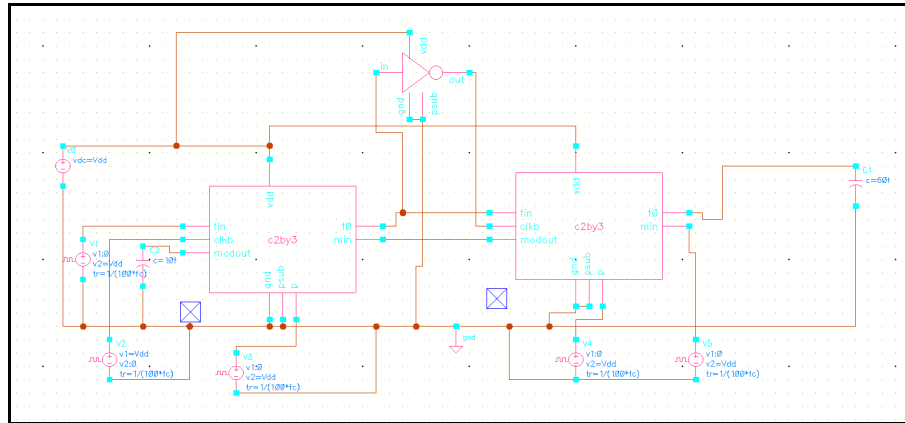


Fig. 3.4.4 Design of 2 cascaded 2/3 divider cell in Cadence
Virtuoso

The below figure contains the timing waveform of the 2/3 divider cell:

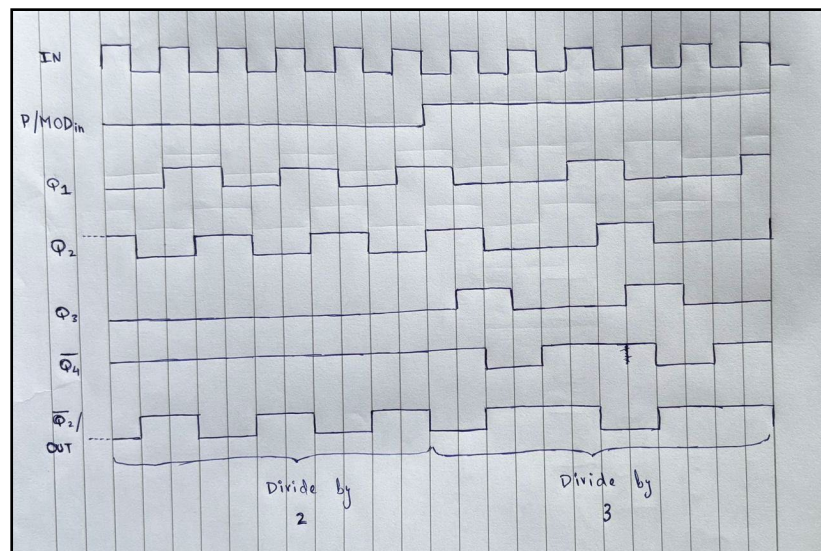


Fig. 3.4.5 Timing waveforms of 2/3 divider cell

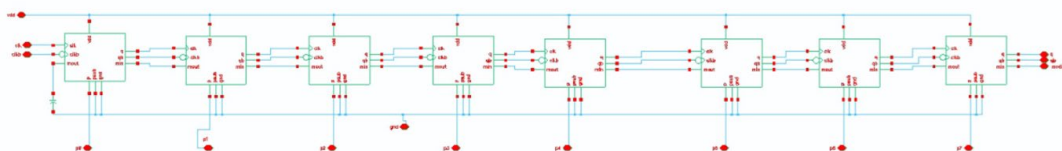


Fig. 3.4.6 Design of 8 cascaded 2/3 divider cell in Cadence Virtuoso

3.5 MASH 111

In a conventional PLL $f_{vco} = N * f_{ref}$. The divide ratio N must be an integer and therefore the resolution of the PLL is f_{ref} . Finer frequency resolution requires a smaller f_{ref} – which causes a narrow loop bandwidth.

In a fractional-N divider, the integer divide ratio is periodically changed from N to $N+1$. The resulting average divide ratio will be increased from N by the duty cycle of $N+1$.

$$\begin{aligned} \text{average } f_{vco} &= \frac{1}{(T_N + T_{N+1})} [T_N * N * f_{ref} \\ &\quad + T_{N+1} * (N + 1) * f_{ref}] \\ &= [N + T_{N+1}/(T_N + T_{N+1})] * f_{ref} \\ &= (N.f) * f_{ref} \end{aligned}$$

Where N is integer part and f is the fractional part of the divide ratio.

The desired fractional frequency is analogous to the analog input of an A2D. Interpolative A2D converters operate by greatly oversampling the input with a coarse 1-bit converter then digitally filtering the 1-bit output to eliminate quantization noise.

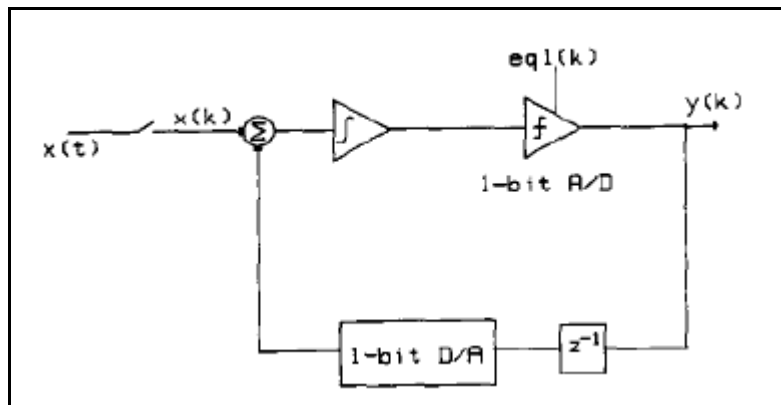


Fig. 3.5.1 Sigma Delta Modulator.

The sigma delta modulator shown in Fig, based on [26] was used in sigma-delta A2D converters. $x(k)$ is the input, $y(k)$ is the output and $eq_1(k)$ is the quantization error introduced by the quantizer. This sigma-delta modulator can be made

suitable for fractional-N synthesis by making the integrator digitally implemented (using an accumulator) and removing the D2A.

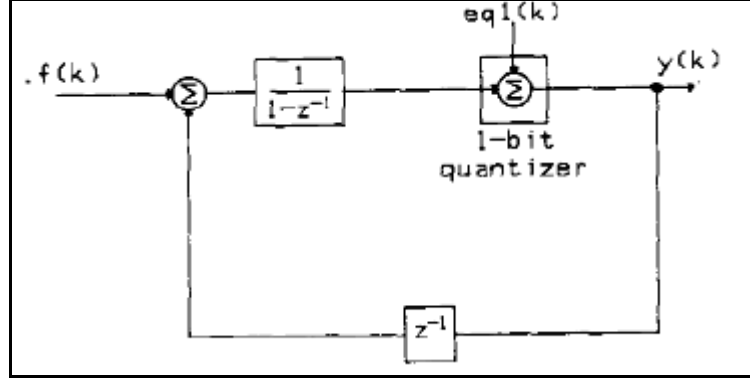


Fig. 3.5.2 Sigma Delta modulator suitable for fractional-N synthesis

$Y(z) = F(z) + (1 - z^{-1})E_{q1}(z)$ from the block diagram.

Three of these modulators are connected to form a fractional-N divider.

$$N_1(z) = (1 - z^{-1})E_{q1}(z) + F(z) \quad (1)$$

$$N_2(z) = -E_{q1}(z) + (1 - z^{-1})E_{q2}(z) \quad (2)$$

$$N'_2(z) = -(1 - z^{-1})E_{q1}(z) + (1 - z^{-1})^2E_{q2}(z) \quad (3)$$

$$N_3(z) = -E_{q2}(z) + (1 - z^{-1})E_{q3}(z) \quad (4)$$

$$N'_3(z) = -(1 - z^{-1})^2E_{q2}(z) + (1 - z^{-1})^3E_{q3}(z) \quad (5)$$

$$\begin{aligned} N_{div}(z) &= N(z) + N_1(z) + N'_2(z) + N'_3(z) \\ &= N.f(z) + (1 - z^{-1})^3E_{q3}(z) \end{aligned} \quad (6)$$

In a locked PLL,

$$f_{out}(k) = N_{div}(k)f_{ref}$$

Using that and (6), we can write

$$F_{out}(z) = N.F(z)f_{ref} + (1 - z^{-1})^3f_{ref}E_{q3}(z) \quad (8)$$

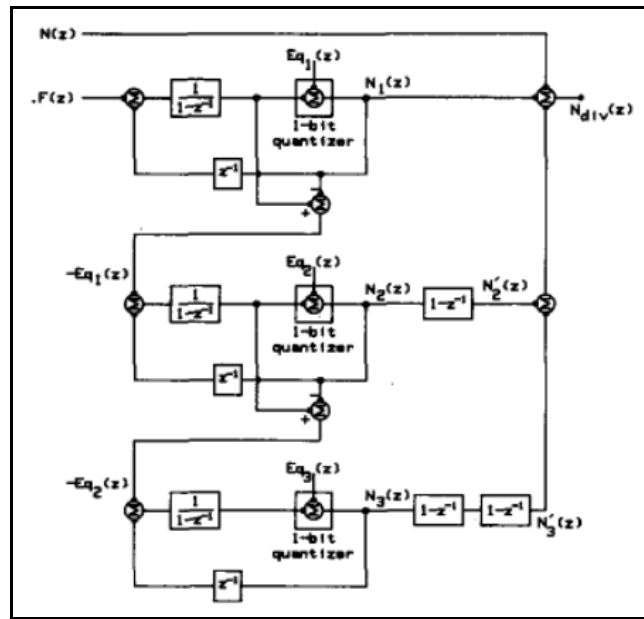


Fig. 3.5.3 Modulator Fractional-N divider.

The structure in Fig can be digitally transformed and represented as an accumulator, like in the following figure:

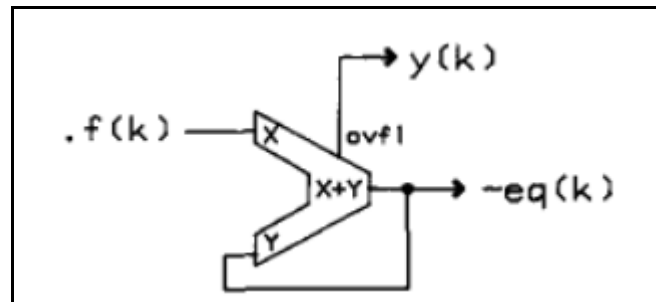


Fig. 3.5.4 Accumulator used as a sigma-delta modulator

The system of Fig. with 3 modulators connected together can be simplified by using an accumulator to represent the sigma-delta modulator in a compact form. Incorporating accumulators in the three-stage sigma-delta modulator yields Fig., which is the system that is currently being used in the Phase Locked Loop.

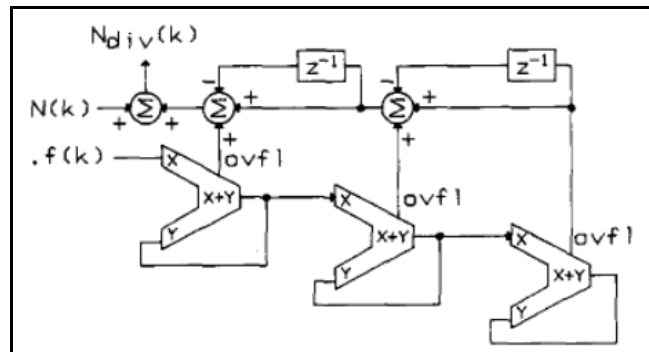
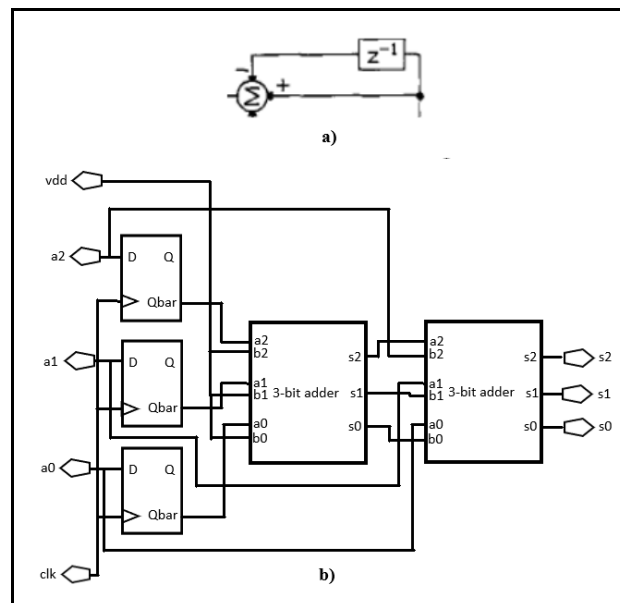


Fig. 3.5.5 MASH 1-1-1 SDM conceptualized using three modulators.

The system is a forward path of accumulators and a reverse path of differentiators. The implementation of the MASH-111 is done by using a fractional resolution of 12-bits (hence, 3 12-bit accumulators have been used) and the reverse path of the differentiators is implemented using a 3-bit compensation network – with differentiators, adders and flipflops. The differentiator implementation is shown in Fig.



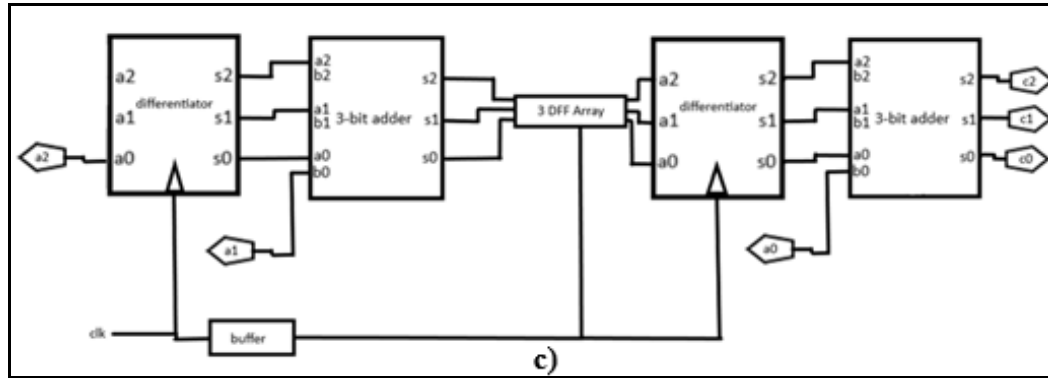


Fig. 3.5.6 (a) Block diagram representation of a differentiator, taken from the compensation network. (b) Digital Implementation of the differentiator. (c) Using the differentiators to design the compensation network.

The Final MASH-111 structure is obtained after combining the accumulators and the compensation network, as shown in Fig. The first accumulator in the series is fed the frequency control word – as previously mentioned in the passage, where $fvco = (N.f) * f_{ref}$, the frequency control word corresponds to $.f$, the fractional part of the divide ratio.

$$.f = \frac{FCW}{no.of\ states}$$

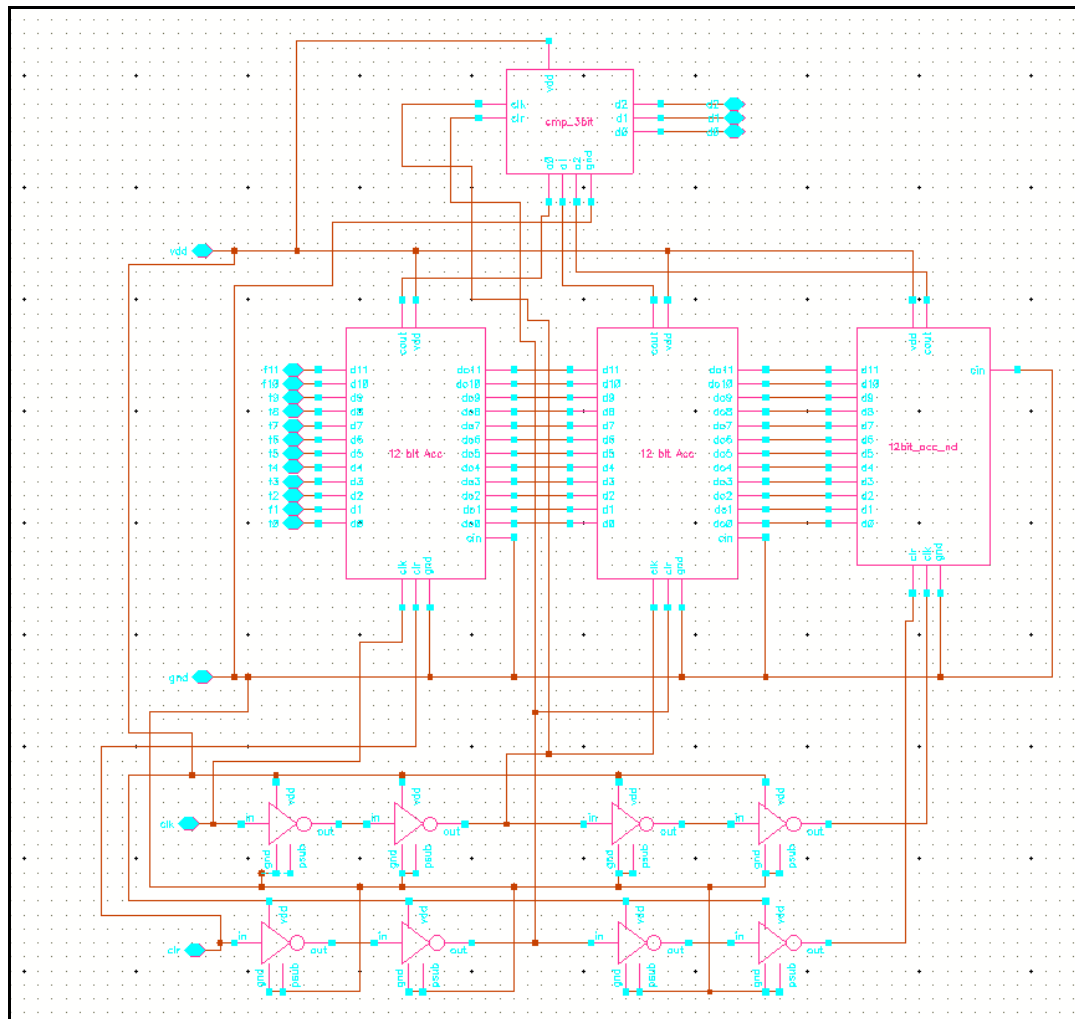


Fig. 3.5.7 Design of the MASH 1-1-1 in Cadence Virtuoso

3.6 PLL

The schematic shown below represents a closed-loop feedback system designed to synthesize a high-frequency, stable output signal from a lower-frequency reference clock. The blocks mentioned above are fitted into the loop to form the closed loop fractional N-PLL system.

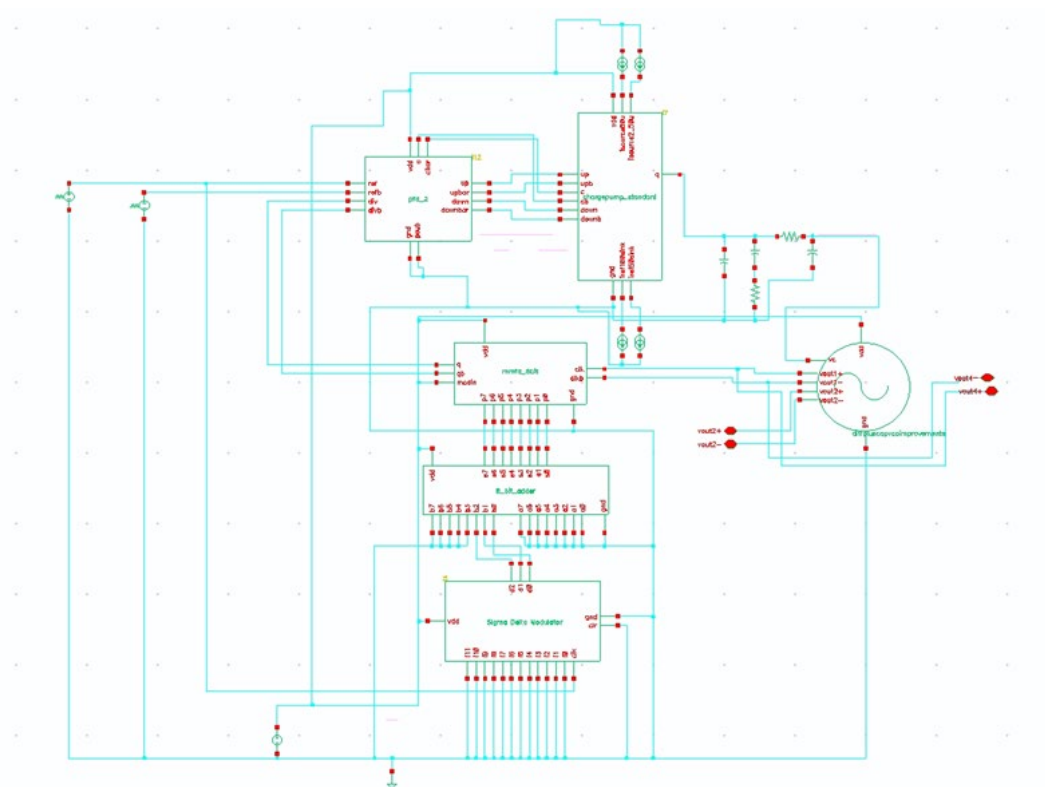


Fig. 3.6.1 Closed loop schematic of fractional N-PLL

CHAPTER 4: RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter includes the results of all the designed blocks used in the PLL - the Phase Frequency Detector, Charge Pump, VCO, MMFD and the Sigma Delta Modulator. It includes the results of all the blocks discussed in the previous chapter. Along with basic transient simulations of all of the blocks, VCO includes phase noise and jitter calculations, and the power consumed, making it ready to be compared with other architectures. The input waveforms or input values are also discussed along with the simulation figure.

4.2 Phase Frequency Detector and Charge Pump



Fig. 4.2.1 PFD and Charge Pump Outputs

The PFD is observed to have fully functioning up and down state pulses. The input condition is given as V_{ref} lagging V_{div} by 3ns, which causes an up-pulse from the PFD. The down state has a slight instance at the end of the up-pulse, to negate any delays, demonstrating the property of a Zero Dead-Zone PFD.

The Charge Pump voltage is observed to be steadily rising as the up-pulse triggers it. This is directly coinciding with the actual functionality of the Charge Pump.

This change in the voltage directly plays an effect on the control voltage of the Voltage Controlled Oscillator, making it change its frequency to lock the Phase Locked Loop to the reference frequency. With up pulses, the charge pump increases its voltage from 1.2 to 1.8, and with down pulses, it decreases the voltage to 0.8.

4.3 Voltage Controlled Oscillator

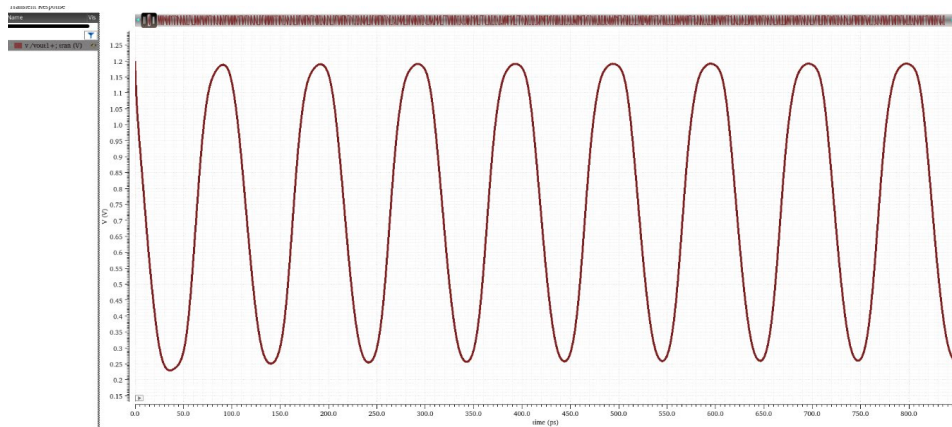


Fig. 4.3.1 Oscillations at $V_{ctrl} = 0$ V achieves a frequency of 9.02GHz

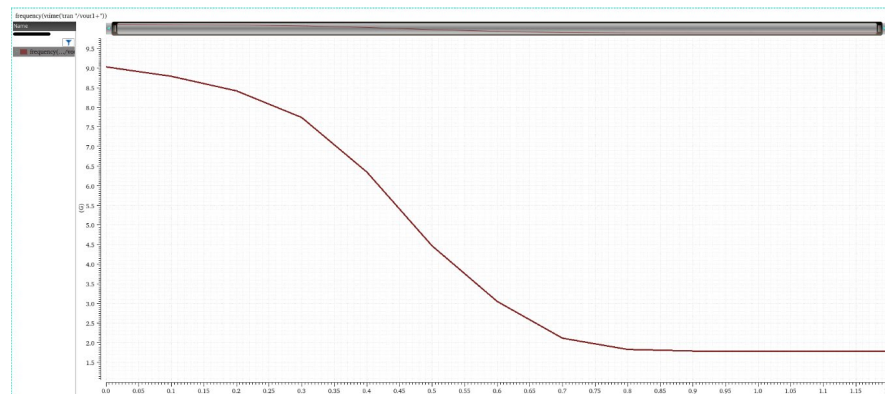


Fig. 4.3.2 Frequency as a function of V_{ctrl} from the range of 0V to V_{dd} V.

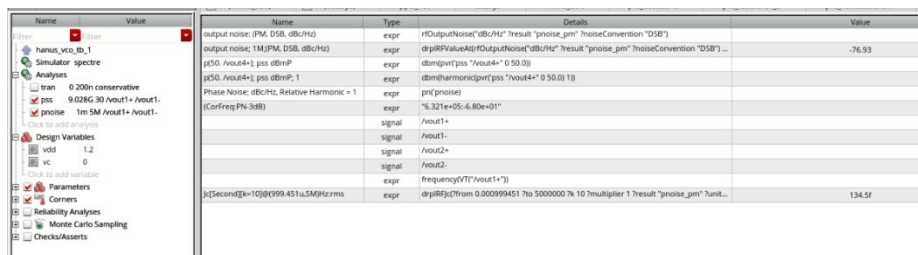


Fig. 4.3.3 Analysis on Maestro along with phase noise and RMS jitter

Thus, the designed Ring-VCO has a wide tuning range from **1.8GHz – 9.02GHz** with control voltages varying between 0 and 1.2 (V_{dd}) V respectively. The following parameters were obtained through the Maestro simulation:

- An RMS jitter of **134.5 fs** at 9.02GHz.
- Phase noise of **-76.93 dbc/Hz** at 1 MHz reference.
- Consumes a Power of **16.35 mW**.

4.4 Multi-Modulus Frequency Divider

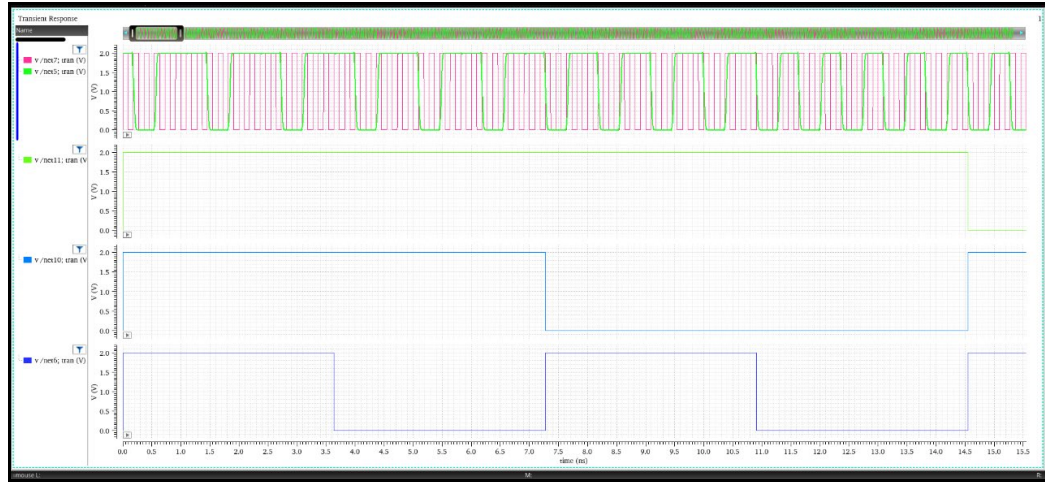


Fig. 4.4.1 MMFD Outputs with varying mod inputs for 2 cascaded 2/3 cells.

The proposed multi-modulus frequency divider successively divides a 5 GHz input clock, with varying divider ratios. For about the first 3.5ns in the simulation, all 3 inputs are on for the cascaded 2/3 cells, which corresponds to Mod_{in}, P₀ and P₁ = 1. When Mod_{in} is high, the divide ratio is:

$$DivideRatio = 2^n + 2^{n-1}P_{n-1} + 2^{n-2}P_{n-2} + \dots + 2^1P_1 + 2^0P_0$$

For cascaded 2/3 cells, with inputs P₀ and P₁, where n = number of cascaded cells, the divide ratio for the first case will be = 4+2+1 =7. This directly coincides with

the divide ratio observed in the accumulator – the output of the MMFD finishes one cycle after nearly 7 cycles of the input clock. Similarly, for the second case, where P_0 is turned off, the divide ratio should be $= 4+2$. The observation matches with this result, as the output clock cycle is finished after 6 cycles of the input clock.

This observation leads to the notion of cascaded more $2/3$ cells to get a larger divide ratio. A larger divide ratio is preferable to lock the PLL to lower, more accurate frequencies and to attain much higher VCO frequencies using the same reference frequency. The MMFD is to be used with the Sigma Delta Modulator to ultimately attain fractional divide ratios to implement the Fractional-N PLL.

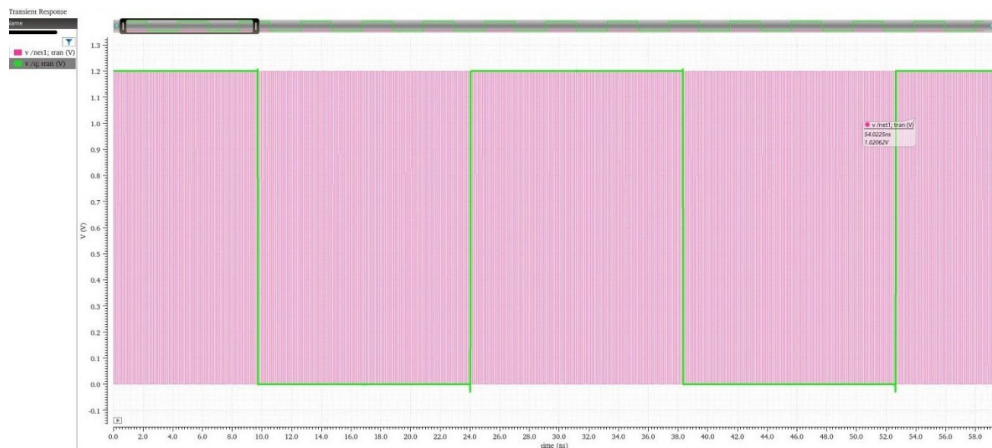


Fig. 4.4.2 MMFD Output with 8 cascaded $2/3$ cells having divide ratio of 256

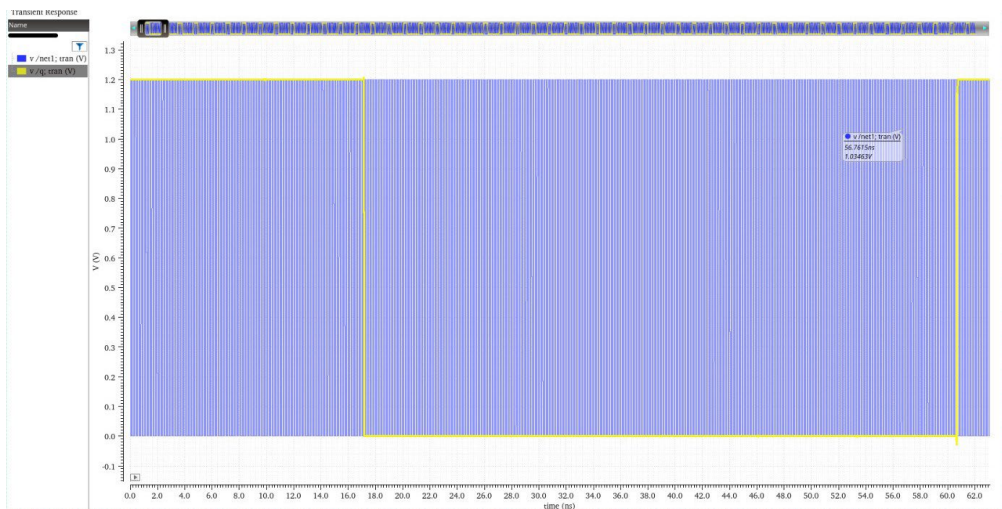


Fig. 4.4.3 MMFD Output with 8 cascaded $2/3$ cells having divide ratio of 511

4.5 Sigma Delta Modulator

Sigma Delta Modulator is fed FCW (Frequency Control Word) of 12 bits. The value is 2319, and the fraction that should be a result from the averages is $3 + (2319/4096) = 3.566$.

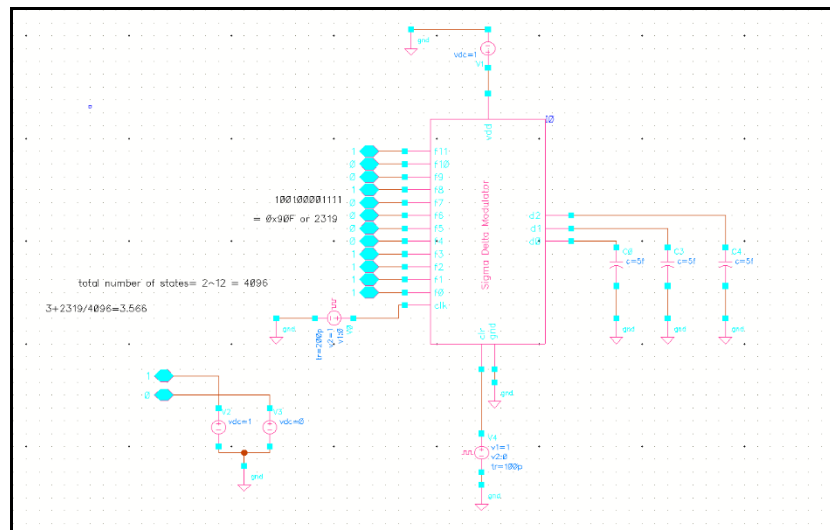


Fig. 4.5.1 Testbench for the Sigma Delta Modulator.

Average is taken from the outputs of the compensation network.

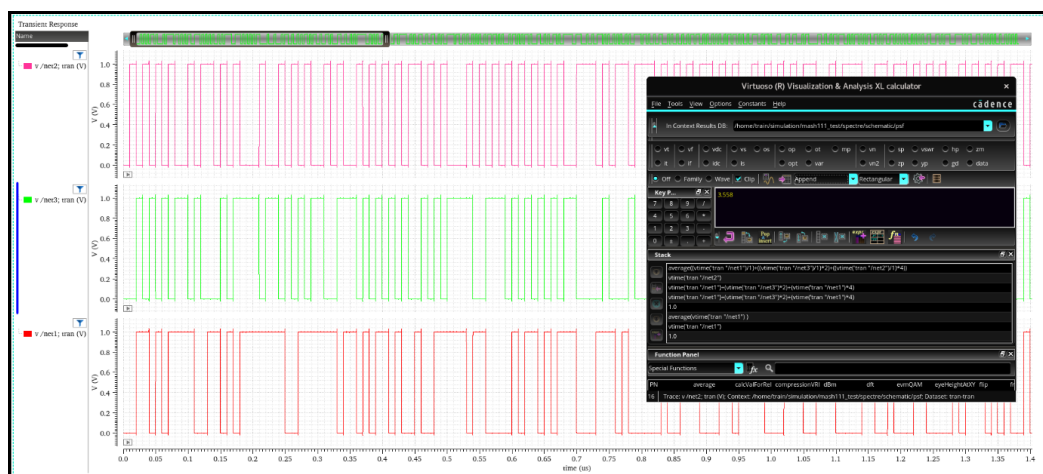


Fig. 4.5.2 Outputs for the Sigma Delta framework

For the calculation of the average from the three outputs, the following formula is used

$$\frac{\sum_{i=0}^{4095} 4o_2^i + 2o_1^i + o_0^i}{4096}$$

Where i refers to the number of clock cycles. Theoretically, it should be a total of 4096 cycles, but transient analysis limits – and processing time might allow much lesser. The average obtained in Fig. is after a transient simulation of 5 μ s, which equates to around 500 clock cycles.

4.6 PLL Results

The Fractional-N Phase Locked Loop is simulated by first setting a few initial voltages and also the division ratio to attain certain output frequencies. The initial voltage for the VCO is set, and the division ratio is set to 256 (for 5.12GHz, lowest frequency possible) and 455.5 (for fractional division and higher frequency test – 9.11GHz). A transient simulation is completed to attain lock time for the PLL.

An adder is used to consolidate the coarse divide ratio and the fine divide ratio into the MMFD.

The waveform result of the transient analysis is given below where each waveform is an output of a particular block from the closed loop fractional N-PLL.

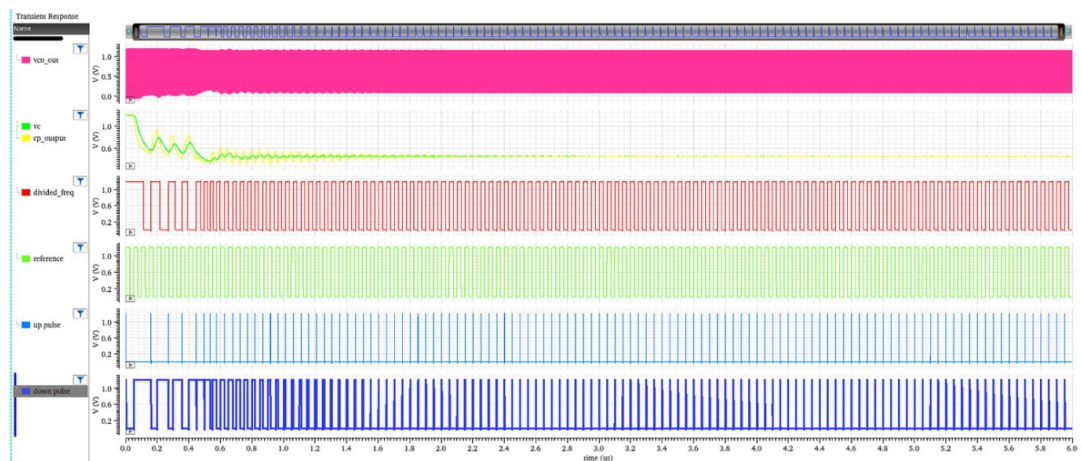


Fig. 4.6.1 Transient waveform simulation of fractional N-PLL

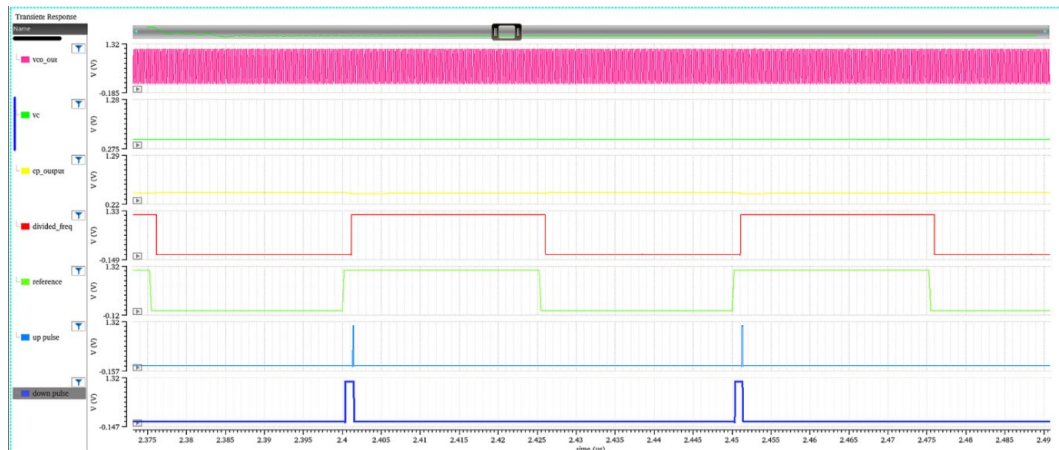


Fig. 4.6.2 Zoomed in waveform

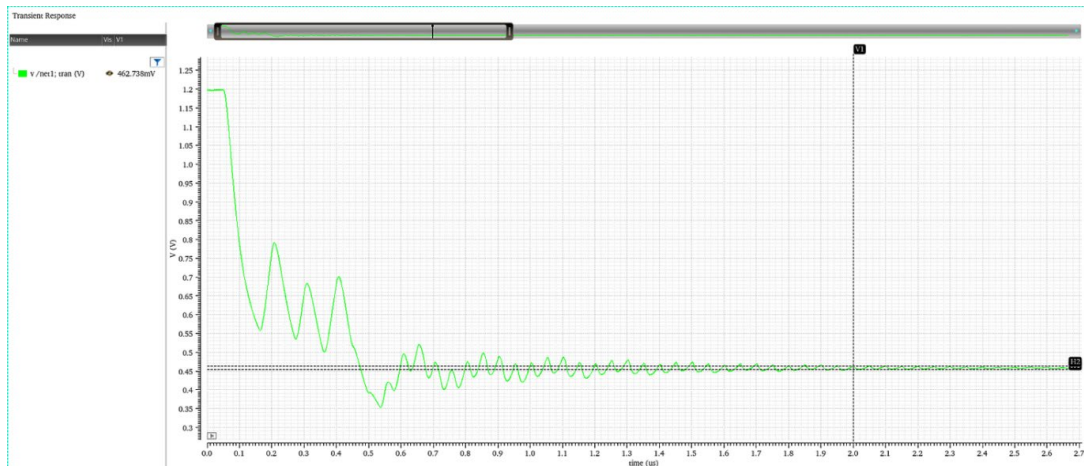


Fig. 4.6.3 Lock time waveform

From the transient simulation results, the closed-loop acquisition and tracking dynamics of the fractional N-PLL can be clearly observed. The loop demonstrates stable settling behaviour, with the voltage control node (V_{ctrl}) entering and remaining within a $\pm 1\%$ error band of its steady-state target value at approximately $t = 2.0\mu s$. This time period is known as the lock time of the PLL. At the onset of the simulation ($t=0s$), V_{ctrl} initializes at the supply rail (V_{dd}). Given the negative tuning sensitivity characteristic of the Voltage-Controlled Oscillator (negative K_{vco}), this maximum control voltage corresponds to the minimum operating frequency of the oscillator. As the negative feedback loop engages, V_{ctrl} is dynamically driven downward, progressively increasing the VCO output frequency toward the target channel. From the fig. 4.6.2, the feedback clock ($divided_freq$) lags the reference by a minor phase delay which

triggers a narrow DOWN pulse from the PFD. This enables the charge pump's sink network to discharge the loop filter and eliminate the phase error by increasing voltage and thus decreasing the frequency. At the end of each DOWN pulse, a small UP pulse is generated because the PFD's internal reset AND gate has a finite propagation delay, keeping both signals momentarily high before clearing. This intentional overlap ensures the pulses are wide enough to fully turn on the charge pump switches, eliminating the dead-zone and preventing phase noise degradation.

The PLL loop consumes a power of **16.35 mW**.

4.7 Comparison table

The table of comparison of the blocks of the fractional N-PLL are given as follows,

FOR MMFD-

Reference	Architecture	Tech	VDD	Max Operating Frequency	Division Ratio	Power
Proposed	MMFD, cascaded 2/3 cells, transmission gate latches	45nm	1.2V	8.95GHz	256-511	2.45mW
[29]	Pulse-swallow MMD using TSPC DFF	90nm	-	9.2GHz	101-164	“Low-power”
[30]	8-stage cascaded 2/3 MMFD	55nm	1V	2GHz	8-511	66.04uW

FOR ZERO DEAD ZONE PFD-

References	Architecture	Tech	VDD	Max Freq	Dead Zone	Power	Area
Proposed	PPCL differential zero-dead-zone PFD	45 nm	1.2 V	—	~0 ps	7.913 μ W	—
[31]	Optimized Symmetric PFD	90 nm	—	5 GHz	25 ps	29 μ W	28.43 μ m ²
[32]	TSPC-based PFD	28 nm	~1 V	3 GHz	40 ps	4.41 μ W	10.42 μ m ²

FOR RING VCO-

References	Architecture	Tech	Supply	Freq	PN	Power	FOM
Proposed	Dual 4-stage differential capacitively coupled ring	45nm	1.2 V	9.028GHz	-76.93 @1MHz	16.35mW	-151.9
[27]	Capacitively phase-coupled quadrature ring (passive IL)	40nm	1.0 V	28.0GHz	-124.5 @10MHz	0.88mW	-194.0
[28]	4-stage differential ring, pseudo-differential feedback	180nm	1.8V	2.08GHz	-93.1 @1MHz	1.89mW	-156.7
[33]	Differential ring VCO (wide tuning)	180nm	-	1.27GHz	-103.1 @1MHz	-	-158.9

FOR PLL-

Reference	Architecture	Tech	VDD	Operating Frequency	Lock Time	Power	Phase Noise	FOM
Proposed	Differential VCO based Fractional PLL	45nm	1.2V	5.12GHz	2.0us	10.9mW	-	-
[34]	Ring-VCO Fractional-N PLL	12nm	0.8V	5.76GHz	-	6.4mW	-96.3 dBc/Hz @1 MHz	-238 dB
[35]	Sub-Sampling PLL	-	0.8V	5.3-5.9GHz	-	-	-	-251.6 dB

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project focuses on the block-level design of a Fractional-N Phase-Locked Loop, emphasizing the selection and implementation of suitable architectures for each functional module. The essential building blocks—including the PFD, charge pump, VCO, MMFD and sigma-delta modulator—were individually developed based on their operational requirements and established design practices. The work carried out provides a clear understanding of the roles and interactions of each component within a Fractional-N synthesis system. The blocks are designed and implemented in the 45nm process technology in Cadence Virtuoso. Phase noise analysis is also done for the VCO architecture used.

5.2 Future Scope

Future work can be done on optimizing the individual blocks for a higher performance. The VCO phase noise is not on par with other structures, because of it being a basic structure. Refinements in the architecture are needed for a better performance. The Power Consumption for the Sigma-Delta Modulator is predicted to be extremely high for the number of transistors used. Implementing different logics for the most basic of cells like the adder, the flip flop may prove to be beneficial for the power consumption. Different architectures of the VCO can also be implemented according to the application. The MMFD can also be designed using different logics like CML and TSPC.

CHAPTER 6: REFERENCES

- [1] Sharanya, K. and Vignesh, V., 2024, July. Charge Pump based Phase Locked Loop with Transmission Gate using Combinational Phase Frequency Detector. In *2024 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT)* (pp. 1-5). IEEE.
- [2] Liu, L., Ji, Y., Liao, X., Qin, Z. and Liang, H., 2022. A 0.8-V, 2.55-GHz, 2.62-mW charge-pump PLL with high spectrum purity. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 30(2), pp.113-122.
- [3] Zouaq, K., Aitoumeri, A., Bouyahyaoui, A. and Alami, M., 2018, November. A novel scheme of high-speed phase-frequency detector for low-power low-phase noise PLL design. In *2018 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)* (pp. 224-229). IEEE.
- [4] Chang, J.R. and Liu, S.I., 2021. A 2–3 GHz fast-locking PLL using phase error compensator. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 69(4), pp.2026-2030.
- [5] Mercandelli, M., Santiccioli, A., Parisi, A., Bertulesi, L., Cherniak, D., Lacaita, A.L., Samori, C. and Levantino, S., 2021. A 12.5-GHz fractional-N type-I sampling PLL achieving 58-fs integrated jitter. *IEEE Journal of Solid-State Circuits*, 57(2), pp.505-517.
- [6] Chou, Y.J., Chen, H.C. and Chang, Y.M., 2023. Design and analysis of sub-sampling phase-locked loop for quantum computing. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 31(9), pp.1330-1338.
- [7] Gong, Y., Yan, W., Luo, M., Wang, C. and Zhang, L., 2023. A frequency-tunable fractional-N PLL for high-energy physics experiments. *IEEE Transactions on Nuclear Science*, 70(4), pp.722-729.
- [8] Shehab, S.A. and Yonis, A.Z., 2024, July. Design and simulation of fractional N-PLL 2.4 GHz for WLAN in IEEE 802.11 standards. In *2024 11th*

- International Conference on Wireless Networks and Mobile Communications (WINCOM)* (pp. 1-6). IEEE.
- [9] Narayanan, A., Bhat, A. and Krishnapura, N., 2024. A 6 to 12-GHz fractional-N frequency synthesizer with a digital technique to counter modulus-dependent feedback divider delays. *IEEE Journal of Solid-State Circuits*, 59(9), pp.2818-2830.
- [10] Lee, Y., Seong, T., Yoo, S. and Choi, J., 2017. A low-jitter and low-reference-spur ring-VCO-based switched-loop filter PLL using a fast phase-error correction technique. *IEEE Journal of Solid-State Circuits*, 53(4), pp.1192-1202.
- [11] Zhang, Z., Zhu, G. and Yue, C.P., 2020. A 0.25–0.4-V, sub-0.11-mW/GHz, 0.15–1.6-GHz PLL using an offset dual-path architecture with dynamic charge pumps. *IEEE Journal of Solid-State Circuits*, 56(6), pp.1871-1885.
- [12] Thakore, K.P., Shah, K. and Devashrey, N.M., 2019, March. Design and implementation of low power phase frequency detector for phase lock loop. In *2019 3rd International Conference on Computing Methodologies and Communication (ICCMC)* (pp. 644-647). IEEE.
- [13] Ahirwar, R., Shankhwar, H.K., Kaushal, G., Pattanaik, M. and Srivastava, P., 2022, December. Design high frequency phase locked loop using single ended VCO for high speed applications. In *2022 IEEE Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI)* (pp. 1-6). IEEE.
- [14] Dutta, D., Tumukunta, S.P., Sivaraaj, N.R. and Majeed, K.A., 2024. Exploring the landscape of phase-locked loop architectures: A comprehensive review. *IEEE Access*.
- [15] Rajalingam, P., Jayakumar, S. and Routray, S., 2021. Design and analysis of low power and high frequency current starved sleep voltage controlled oscillator for phase locked loop application. *Silicon*, 13(8), pp.2715-2726.
- [16] Sivasakthi, M. and Radhika, P., 2024. Design and analysis of PVT tolerant hybrid current starved ring VCO with bulk driven keeper technique at 45 nm

- CMOS technology for the PLL application. *AEU-International Journal of Electronics and Communications*, 173, p.154987.
- [17] Gong, H., Liao, X., Li, Y., & Zhao, D. (2024, October). A 11.4-GHz Dual-Mode Multi-Modulus Divider in 65-nm CMOS. In *2024 IEEE International Conference on Integrated Circuits, Technologies and Applications (ICTA)* (pp. 25-26). IEEE.
- [18] Rai, A., Kundu, K., & Dev, R. (2023, April). Analysis of Phase Frequency Detector & Charge Pump Switching Circuits for Frequency Synthesizer Using CMOS Technology. In *2023 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES)* (pp. 605-609). IEEE.
- [19] Sallah, S. S. M., Azulyzal, A. I. H., Noorsal, E., Abd Rashid, A. N., & Ahmed, A. S. A. (2024, July). Optimization of a Current-Starved Ring-Based Voltage Controlled Oscillator (VCO) for High-Frequency Band Phase Locked Loop (PLL) using 45nm CMOS Technology. In *2024 IEEE International Conference on Applied Electronics and Engineering (ICAEE)* (pp. 1-5). IEEE.
- [20] Kinger, B., Suman, S., Sharma, K. G., & Ghosh, P. K. (2015, February). Design of improved performance voltage controlled ring oscillator. In *2015 Fifth International Conference on Advanced Computing & Communication Technologies* (pp. 441-445). IEEE.
- [21] Osada, M., Xu, Z., & Iizuka, T. (2022, September). An inductorless fractional-N PLL using harmonic-mixer-based dual feedback and high-OSR delta-sigma-modulator with phase-domain filtering. In *ESSCIRC 2022-IEEE 48th European Solid State Circuits Conference (ESSCIRC)* (pp. 245-248). IEEE.
- [22] Zhong, J., Yang, X., Martins, R. P., Zhu, Y., & Chan, C. H. (2023). A 0.016 mm² Active Area 4GHz Fully Ring-Oscillator-Based Cascaded Fractional-N PLL With Burst-Mode Sampling. *IEEE Transactions on Circuits and Systems II: Express Briefs*, 70(10), 3792-3796.
- [23] Hong, N. P., Nam, K. H., & Park, J. S. (2022). A fractional-N synthesizer based on programmable frequency multiplier for 5G+ communication system. *IEEE Transactions on Microwave Theory and Techniques*, 71(4), 1518-1535.

- [24] Gong, Y., Yan, W., Luo, M., Wang, C., & Zhang, L. (2023). A frequency-tunable fractional-N PLL for high-energy physics experiments. *IEEE Transactions on Nuclear Science*, 70(4), 722-729.
- [25] Feng, L., Rhee, W., & Wang, Z. (2023). A DTC-free fractional-N BBPLL with FIR-embedded injection-locked-oscillator-based phase-domain lowpass filter. *IEEE Journal of Solid-State Circuits*, 59(3), 728-739.
- [26] Miller, B., & Conley, B. (1990, May). A multiple modulator fractional divider. In *44th annual symposium on frequency control* (pp. 559-568). IEEE.
- [27] Schober, S.M. and Choma, J., 2015, June. A capacitively phase-coupled low noise, low power 0.8-to-28.2 GHz quadrature ring VCO in 40nm CMOS. In *2015 IEEE 13th International New Circuits and Systems Conference (NEWCAS)* (pp. 1-4). IEEE.
- [28] Ding, Z., Zhang, J., Guo, A. and Yin, L., 2024, October. Design of a Novel Low-Power Differential Ring Voltage Controlled Oscillator. In *2024 9th International Conference on Integrated Circuits and Microsystems (ICICM)* (pp. 473-478). IEEE.
- [29] Zhao, Y., Chen, Z., Liu, Z., Li, X. and Wang, X., 2020. A 4.1 GHz–9.2 GHz programmable frequency divider for Ka Band pll frequency synthesizer. *Electronics*, 9(11), p.1773.
- [30] Zhao, M. and Huang, J., 2023, June. Design of a multi-modulus divider with a wide frequency dividing range and low power consumption. In *Journal of Physics: Conference Series* (Vol. 2524, No. 1, p. 012008). IOP Publishing.
- [31] Hemel, M.S.K., Reaz, M.B.I., Ali, S.H.B.M., Bhuiyan, M.A.S. and Miraz, M.H., 2024. Optimisation and performance computation of a phase frequency detector module for IoT devices. *arXiv preprint arXiv:2401.03389*.
- [32] Challagundla, D., Sundarapu, V.K.V., Bezzam, I. and Islam, R., 2025, October. TSPC-Based Low-Power High-Resolution CMOS Phase Frequency Detector. In *2025 IFIP/IEEE 33rd International Conference on Very Large Scale Integration (VLSI-SoC)* (pp. 1-5). IEEE.
- [33] Kumar, N. and Kumar, M., 2021. Low power wide tuning range differential ring VCO designs with I-MOS and A-MOS varactor. *AEU-International Journal of Electronics and Communications*, 131, p.153583.

- [34] Li, Y., Xun, B., Shi, Y., Xu, X., Li, M., Zhu, H. and Sun, Q., 2024. A Compact, Low-Power, and Low-Jitter Fractional-N Phase-Locked Loop with a Single-Ended Ring Voltage-Controlled Oscillator in a 12 nm CMOS FinFET. *Electronics*, 13(13), p.2617.
- [35] Zuo, S., Zhao, J. and Zhou, Y., 2021. A 0.8 V, 5.3–5.9 GHz Sub-Sampling PLL with 196.5 fs rms Integrated Jitter and– 251.6 dB FoM. *Sensors*, 21(22), p.7648.